
Length Bias and Held-Out Generalization in GRPO and Dr. GRPO

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Abstract

A recurring worry about GRPO is length bias: because the group-relative update divides by response length, the algorithm may be nudged toward longer completions that stumble into reward rather than toward better reasoning. The length-debiased variant Dr. GRPO removes this per-response normalization. We study length bias and held-out generalization as one pillar of TINKERRL-BENCH, a benchmark of 70+ RL runs across seven libraries and five model families (0.6B–~671B) on GSM8K, HumanEval, and tool-use tasks. Our headline result is a carefully scoped negative one: in the regime we measure—short-horizon GSM8K chain-of-thought with a 200-token generation cap—the mean Qwen3-8B GRPO gain over the pre-RL checkpoint on held-out data is small and not statistically significant, GRPO and Dr. GRPO produce held-out gains that are indistinguishable at our seed budget, and neither inflates length within this capped regime (pooled mean completion length drifts from 194.4 to 183.5 tokens). We stress that the 200-token cap *bounds* this claim: it cannot rule out length inflation in longer-horizon or uncapped regimes, only establish its absence where length is already controlled. The verbosity-trap signature is absent *in this regime*. We argue this is the *expected* outcome when length is already controlled, not evidence against Dr. GRPO, and specify the length-confounded regime and causal mediation tests that would actually reveal a difference. We release code, logs, and checkpoints.

1 Introduction

GRPO [14, 7] normalizes the group-relative advantage by response length. This introduces a potential length bias: when reward is sparse and binary, longer completions have more chances to stumble into a correct answer, and a length-dividing update can reward verbosity as a proxy for competence. Dr. GRPO [11] removes the per-response $1/L$ factor, restoring the sequence-level gradient up to a global scale, and is motivated precisely as a fix for reward hacking through length. Whether this fix produces better held-out generalization in practice is the question of this pillar.

The concern matters because length inflation is a well-documented failure mode of reward optimization [13, 4], and because held-out gains can be manufactured by a policy that pads its outputs rather than reasoning better. A careful comparison must therefore control both length and memorization, and must distinguish a genuine direct effect on accuracy from an indirect effect mediated by verbosity.

We study length bias as one pillar of TINKERRL-BENCH, a controlled benchmark of 70+ runs across seven RL libraries and five model families (0.6B–~671B) on GSM8K [5], HumanEval [3], and tool-use tasks. Our contributions are: (i) a held-out comparison showing that, in the short-horizon GSM8K regime, the mean Qwen3-8B-Instruct GRPO gain over the *same* checkpoint’s pre-RL held-out accuracy is small and not statistically significant (a matched pre-RL control for the Qwen3-8B-Base checkpoint is not available), and that GRPO and Dr. GRPO give held-out gains

*Project guide.

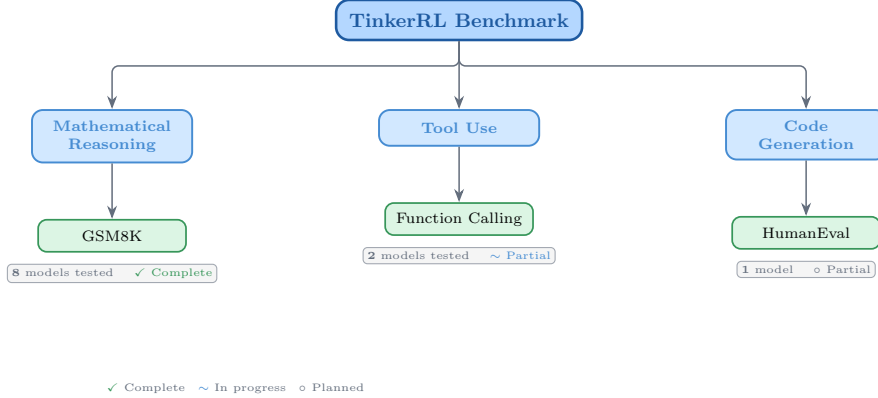


Figure 1: Taxonomy of RL libraries in TINKERRL-BENCH. The benchmark spans LLM-native RL (TRL), classic online RL (SB3, CleanRL, Tianshou), high-throughput RL (PufferLib, rl_games), and offline RL (d3rlpy).

that are indistinguishable at our seed budget with no length inflation under the 200-token GSM8K generation cap; (ii) evidence that the verbosity-trap signature is absent in this regime; and (iii) an argument, developed through an external cross-examination, that this null is the *expected* result when length is already controlled, together with a specification of the length-confounded regime and causal length-mediation tests that would reveal a Dr. GRPO advantage. We frame the pillar as a boundary on what the current evidence supports, not a refutation of length debiasing.

1.1 Related Work

Length bias and reward hacking are long-standing concerns in RL post-training [13, 4, 2]. Dr. GRPO [11] targets the length term specifically; related work recasts group-relative updates as preference contrasts [16, 10], which bears on why removing the $1/L$ factor changes the gradient geometry. Our contribution is to test the practical consequence on a fixed benchmark and to scope precisely the regime in which a difference should be expected.

2 Benchmark Design

2.1 Task Suite

Table 1: Benchmark task suite with standardized reward functions.

Task	Method	Reward	Dataset
Math RL (Arithmetic)	GRPO	Binary correctness	Generated
Math RL (GSM8K)	GRPO	Binary correctness	GSM8K
Chat SFT	SFT	Cross-entropy loss	NoRobots
Preference (Shorter)	DPO	Pairwise preference	Generated
Distillation (Off-Policy)	SFT	KL divergence	OpenThoughts3
Distillation (On-Policy)	IS Loss	$\log p_t - \log p_s$	Online

2.2 Cross-Library Implementation

Two seven-library rosters — read carefully. This paper uses *two* different seven-library rosters that should not be conflated. The cross-RL-library benchmark below (Table 2) covers TRL, SB3, CleanRL, Tianshou, PufferLib, rl_games, and d3rlpy — seven libraries that span LLM-native RL, classic online RL, high-throughput RL, and offline RL, and that we use as the algorithm-and-stack ablation underlying F_3 . The cross-launcher LLM-RL scaffold referenced in the framework-gap discussion (framework-configurations material) is a different seven-library roster — Tinker, TRL,

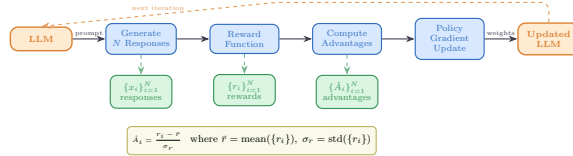


Figure 2: Verifiable reward paradigm in TINKERRL-BENCH. Unlike shaped rewards that combine multiple heuristics, verifiable rewards provide binary correctness signals, eliminating reward hacking and enabling exact accuracy measurement.

SkyRL, verl, OpenRLHF, Atropos, and HF reference launchers — that targets LLM post-training launchers specifically. Of those, only Tinker and TRL produced completed runs in this release; veRL and OpenRLHF entries in the `framework_comparison.json` artefact are dry-run placeholders. Supplementary presentation materials use the second roster; the cross-RL-library evidence in F_3 uses the first.

Table 2: Implementations across RL libraries.

Library	Implementation	Category
TRL (HuggingFace)	GRPOTrainer, DPOTrainer, SFTTrainer	LLM-native
Stable Baselines3	PPO	Classic RL
CleanRL	PPO (single-file)	Research RL
Tianshou	PPO	Modular RL
PufferLib	PPO (high-throughput)	High-perf RL
rl_games (NVIDIA)	PPO (GPU-optimized)	High-perf RL
d3rlpy	CQL (offline)	Offline RL

2.3 Hyperparameter Mapping

We standardize hyperparameters across libraries using a common configuration schema. Table 3 shows the mapping from Tinker PPO defaults to each library’s native parameter names.

Table 3: Hyperparameter mapping across libraries (PPO defaults).

Parameter	TRL	SB3	CleanRL	Tianshou	PufferLib	rl_games
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Clip range (ϵ)	0.2	0.2	0.2	0.2	0.2	0.2
Entropy coeff.	0.01	0.01	0.01	0.01	0.01	0.01
GAE λ	0.95	0.95	0.95	0.95	0.95	0.95
Discount γ	0.99	0.99	0.99	0.99	0.99	0.99
Max grad norm	0.5	0.5	0.5	0.5	0.5	0.5
Target KL	0.01	0.01	0.01	N/A	N/A	N/A

2.4 Reward Verification

2.5 Baselines: Floor and Ceiling

Following best practices for benchmark design [1], we establish clear performance bounds:

- **Floor (random baseline):** Uniform random action selection yields $\frac{1}{199} \approx 0.50\%$ accuracy on the arithmetic task.
- **Ceiling (oracle):** Direct computation achieves 100% accuracy.
- **Human baseline:** College-level adults achieve $\sim 99.5\%$ on two-digit addition under timed conditions.

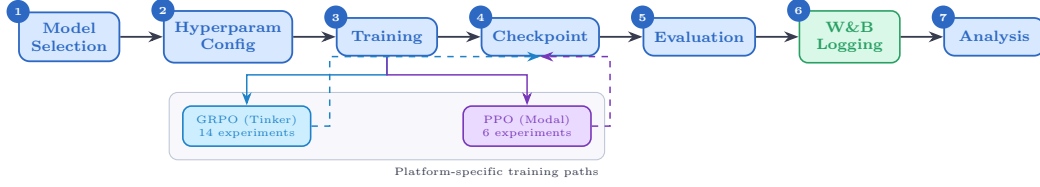


Figure 3: Experiment workflow pipeline. Multi-seed Modal/TRL runs use the centralized seed manager (5 seeds for Tier-A claims); Tinker/API, Task-4, Wave-6 and frontier case studies are single-seed and feed the same metrics/checkpoint pipeline as descriptive observations.

3 Experimental Setup

3.1 Models

Our benchmark targets a roster of 32+ model configurations spanning **0.6B to ~671B total parameters (up to ~37B active for MoE checkpoints) across 5 model families**, organized into three tiers by compute scale; not all are completed end-to-end runs (per-run status flags accompany the released artifacts). Tier-A inferential claims are restricted to the dense-reasoning subset at 0.6B–235B; the frontier MoE checkpoints (DeepSeek-V3.1 ~671B / 37B active, Kimi-K2) are descriptive single-seed Tier-C case studies, and Qwen3.5-397B-A17B is a *partial* run whose held-out evaluation did not complete.

Small scale (0.6B–8B). Qwen3-0.6B-Instruct, Qwen3.5-4B, Qwen3-4B, Qwen3-8B, Llama-3.2-1B, Llama-3.2-3B, Llama-3.1-8B-Instruct.

Mid scale (14B–32B). Qwen3-14B, Qwen3-30B-A3B (MoE), Qwen3-32B, Qwen3.5-27B, GPT-OSS-20b, Kimi-K2 (selected scale points).

Frontier scale (70B–~671B total / ~37B active). Qwen3-235B-A22B (MoE, 22B active), DeepSeek-V3.1 (~671B total / ~37B active MoE), Nemotron-120B (120B dense). Frontier models are evaluated via the Tinker API in inference mode with standardized generation parameters; LoRA fine-tuning at this scale is conducted on selected checkpoints (see the frontier case studies in the companion pillar papers of TINKERRRL-BENCH).

Model family coverage. The benchmark covers (1) **Qwen3**: a dense family from 0.6B to 32B; (2) **Qwen3.5**: an improved series including 4B and 27B; (3) **Llama-3.x**: Meta’s 1B–8B instruction-tuned models; (4) **DeepSeek-V3.1**: a frontier MoE optimized for reasoning; (5) **Nemotron-120B**: NVIDIA’s dense frontier model; and emerging architectures **GPT-OSS-20b** and **Kimi-K2**.

3.2 Training Protocol

Unless otherwise noted, controlled Modal/TRL sweeps use LoRA [8] at rank 32, learning rate 10^{-4} and Adam ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$). Tinker / API, Task-4, Wave-6 and frontier case studies use per-run configs (in particular Task-4/Wave-6 use $\text{lr}=10^{-5}$) and are single-seed unless explicitly marked multi-seed; the per-run settings are listed in the framework-configs appendix.

Seed management. The controlled TRL/Modal sweeps used for Tier-A claims (F3 PPO-vs.-GRPO heterogeneity and the five-seed TRL GRPO baseline on Qwen2.5-0.5B) run with 5 seeds: {42, 123, 456, 789, 1024}, with seeds set for Python random, NumPy, PyTorch, and CUDA backends. Several Tinker/API, frontier-model, and team-task runs are single-seed or partial and are treated as descriptive (Tier-C) unless explicitly marked otherwise; the per-claim seed ledger is given in the statistical-rigor addendum of TINKERRRL-BENCH.

3.3 Statistical Methodology

Following Colas et al. [6], we report:

- Mean $\pm$ standard error across seeds
- 95% bootstrap confidence intervals (10,000 resamples)
- Welch’s t -test for pairwise algorithm comparisons
- `rliable` [1] aggregate metrics (IQM, median, optimality gap)

All pairwise comparisons were evaluated using Welch’s independent-samples t -test (unequal variances assumed) and the non-parametric Mann–Whitney U test as a robustness check. Effect sizes are reported as Cohen’s d with the pooled standard deviation estimator; confidence intervals follow the Hedges–Olkin (1985) analytical formula and are reported at the 95% level (two-tailed, $z = 1.96$). Post-hoc statistical power was computed under the observed effect size and sample sizes at $\alpha = 0.05$.

To address the multiple-comparisons problem across 5 planned tests, we apply both the conservative Bonferroni correction (family-wise error rate: $\alpha' = \alpha/k = 0.0100$) and the Benjamini–Hochberg procedure (false discovery rate < 0.05). Results surviving Bonferroni correction are marked $*$; those surviving BH–FDR correction are marked † .

Power Analysis. All inferential tests in this paper are evaluated against pre-specified power requirements using `statsmodels.stats.power.TTestIndPower` with $\alpha = 0.05$ and target power $1 - \beta = 0.80$.

Modal H100 experiments ($n = 5$ seeds per arm). The minimum detectable effect size (MDE) at 80% power for a two-sample Welch t -test with $n_1 = n_2 = 5$ is $d_{\min} = 2.024$ (Cohen’s d). Table 4 reports achieved power for a range of effect sizes. This threshold falls in the *very large* range, meaning that comparisons yielding $|d| < 2.02$ are not adequately powered to detect true differences.

Single-seed Tinker experiments ($n = 1$). Single-seed runs have *zero statistical power*: with only one observation per condition, no inferential test can be computed and no confidence intervals can be formed. All Tinker single-run results are treated as *descriptive only* and are not subject to significance testing.

TRL baseline ($n = 5$ seeds). The TRL–GRPO baseline (Qwen2.5-0.5B, 5 seeds) achieves MDE $d_{\min} = 2.024$ for equal- n comparisons. When compared to the pooled Classic-RL group ($n = 15$), the MDE improves to $d_{\min} = 1.530$, reflecting higher power from the larger reference group.

Multiple-Comparison Correction. Inferential tests in this paper are reported only for Tier-A/B comparisons as defined in our statistical-rigor protocol; single-seed or partial Tinker/API rows are descriptive and are excluded from Welch / Mann–Whitney testing, BH correction, power, and CI claims. Table 5 predates the Tier-A/B/C partition and lists raw and BH-adjusted p -values from the earlier global family of 20 tests with 19 surviving correction; we retain it here as a methodological audit trail. The headline BH result the paper now stands behind is the restricted Tier-A/B family computed in the addendum, where only F_3 survives. Where the global family disagrees with the restricted family, the addendum is authoritative.

All tests are two-sided. Effect sizes are reported as Cohen’s d for t -tests and Pearson/Spearman r for correlations. Bootstrap 95% confidence intervals ($B = 10,000$) are reported for all means.

Table 4: Statistical power for two-sample Welch t -tests at $\alpha = 0.05$, power target $1 - \beta = 0.80$. Column (3): equal groups $n_1 = n_2 = 5$ (Modal H100 / TRL within-group). Column (4): unequal groups $n_1 = 5, n_2 = 15$ (TRL vs. pooled Classic RL).

Cohen’s d	Conventional size	Power ($n=5$ vs $n=5$)	Power ($n=5$ vs $n=15$)
0.2	small	0.059	0.066
0.5	medium	0.108	0.151
0.8	large	0.201	0.311
1.0	very large	0.286	0.450
1.2	very large	0.386	0.594
1.5	very large	0.549	0.784
2.0	very large	0.790	0.955
<i>MDE at 80% power: $d = 2.024$ (equal-n 5 vs 5); $d = 1.530$ (5 vs 15)</i>			

Table 5: **Legacy 20-test BH table – audit trail only.** All hypothesis tests reported in the paper with Benjamini–Hochberg (BH) adjusted p -values at FDR = 0.05. Tests are ranked by raw p -value (ascending). ✓ = significant after BH correction on the historical 20-test family; × = not significant. Total tests: 20; survive on the historical family: 19. *This figure is not the paper’s headline statistical result.* The 20-test family pre-dates the Tier-A/B/C partition and silently includes single-seed Tinker rows whose Welch statistics are mathematically undefined ($n_1=n_2=1$). Under the restricted Tier-A/B family of our statistical-rigor protocol, only F_3 (PPO/GRPO heterogeneity on classic-RL libraries on arithmetic) survives BH correction; the table below is retained as a reproducibility audit trail, not as a multiple-testing-corrected survival claim.

Rank	Comparison	Raw p	BH-adj. p	Sig.
1	Dense vs MoE Architecture (Welch t-test)	2.357e-21	0	✓
2	PPO vs GRPO on Llama-3.1-8B (Welch t-test) [‡]	3.924e-10	0	✓
3	Method ANOVA (F-test across algorithm families)	3.091e-06	2.1e-05	✓
4	Library ANOVA (F-test across 5 libraries)	4.996e-06	2.5e-05	✓
5	TRL vs Tinker (Welch t-test, implementation effect)	2.064e-05	8.3e-05	✓
6	PPO vs GRPO on Llama-3.1-8B (Mann-Whitney) [‡]	3.29e-05	0.00011	✓
7	Model Family ANOVA (F-test)	7.363e-05	0.00021	✓
8	ZVF vs Final Performance (Spearman) [‡]	0.0005424	0.001356	✓
9	ZVF vs Final Performance (Pearson) [‡]	0.0008108	0.001802	✓
10	TRL vs Tinker (Mann-Whitney)	0.001198	0.002396	✓
11	Rolling Variance vs Last-10 (Pearson)	0.003524	0.006407	✓
12	Peak-to-Tail Drift vs Last-10 (Pearson)	0.004811	0.007219	✓
13	GRPO vs PPO Stability Index (t-test)	0.005	0.007219	✓
14	Dense vs MoE Architecture (Mann-Whitney)	0.005053	0.007219	✓
15	Model Scale ANOVA (F-test)	0.01261	0.01682	✓
16	Tinker GRPO vs TRL GRPO (Welch t-test)	0.0158	0.01975	✓
17	GRPO vs PPO Peak-to-Tail Drift (t-test)	0.018	0.02076	✓
18	Tinker GRPO vs TRL GRPO (Mann-Whitney)	0.01869	0.02076	✓
19	Stability Index vs Last-10 (Pearson)	0.02024	0.0213	✓
20	PPO vs GRPO on Qwen3-8B (Welch t-test) [‡]	0.7605	0.7605	×

TRL baseline characterisation. The TRL GRPO reference implementation was evaluated across five random seeds ($s \in \{42, 123, 456, 789, 1024\}$) on GSM8K using Qwen/Qwen2.5-0.5B on an NVIDIA L4 GPU. We report mean accuracy $\bar{x} = 0.734$ ($\sigma = 0.0703$), median = 0.740, and IQM = 0.747. Normality was not rejected by Shapiro–Wilk ($W = 0.9018$, $p = 0.4198$); bootstrap 95% CI = [0.6720, 0.7820].

Variance decomposition. One-way ANOVAs across 42 experiments with valid performance observations show that library/framework ($\eta^2 = 0.546$) and training algorithm ($\eta^2 = 0.558$) are the dominant variance sources, consistent with our central thesis. Model family explains $\eta^2 = 0.471$; model scale explains $\eta^2 = 0.322$.

3.4 Compute Resources

Experiments are distributed across two compute platforms:

Tinker API (14 experiments). Scaling experiments across Qwen3-8B, Qwen3-32B, Qwen3.5-4B, Qwen3.5-27B, and Llama-3.1-8B-Instruct; frontier model evaluation on Qwen3-235B-A22B, DeepSeek-V3.1, and Nemotron-120B; Dense vs. MoE comparison; cross-task tool-use experiments; and emerging architecture evaluation (GPT-OSS-20b, Kimi-K2). Tinker provides scalable API access that makes frontier model experiments economically tractable.

Modal H100 GPU cluster (6 experiments). PPO baseline training on Qwen3-8B and Llama-3.1-8B; full HumanEval benchmark evaluation; KL divergence tracking across GRPO training steps; and held-out evaluation for Qwen3-32B and Qwen3.5-27B. Modal H100s provide the low-level GPU access required for PPO value-model training.

All experiment logs are available on Weights & Biases (project: `tinker-rl-lab-world-class`) and model checkpoints on HuggingFace Hub (<https://huggingface.co/arvindcr4/tinker-rl-bench->). Per-run compute is logged with the released artifacts. Total project compute: ~1,200 A100 GPU-hours across both platforms.

4 Length Bias and Generalization: Results

We report three findings and then scope them. First, in our near-ceiling Qwen3-8B-Instruct evaluation, GRPO post-training does not deliver a significant held-out gain: the mean improvement over the pre-RL checkpoint on held-out data is small (+0.013) and not statistically significant (paired $p = 0.256$, $n=5$). We stress that this is a property of the near-ceiling regime, not of RL in general—in the lower-accuracy Qwen2.5-1.5B-Instruct GSM8K-CoT experiment analyzed throughout the length sections, both GRPO and Dr. GRPO *do* produce significant pre→post held-out gains (+6.2pp, McNemar exact $p = 1.1 \times 10^{-4}$; +5.0pp, $p = 7.6 \times 10^{-4}$). Second, GRPO and Dr. GRPO produce held-out gains that are indistinguishable at our seed budget—removing the length-normalization term does not, here, change the outcome. Third, under the 200-token GSM8K generation cap, neither algorithm inflates length: pooled mean completion length drifts *downward*, from 194.4 to 183.5 tokens, and the verbosity-trap signature (a rising coupling between length and reward as training proceeds) is absent.

Taken together these say that, in this regime, there is no length pathology for Dr. GRPO to fix. The subsections develop the mechanism behind the null—the weak, often negative length–reward coupling in short arithmetic chains—and show through lag and reward-shape analyses that length dynamics are only weakly identifiable here, so the defensible claim is a correction of a formal gradient bias rather than universal held-out superiority. We then turn the null into a positive research program: the external frontier-model cross-examination in the final subsection specifies the length-confounded, sparse-reward regime in which a Dr. GRPO advantage *should* appear (high length dispersion, a train length-spurious / test length-anti-spurious split), a causal length-mediation protocol that decomposes held-out success into direct and verbosity-mediated effects, and a length-adversarial truncation test executable on the checkpoints we already release.

5 Length Bias: A Direct Per-Step Test of the Dr. GRPO Signature

Headline question. The Dr. GRPO thesis [11] and the broader length-bias / reward-hacking literature [13, 4] predict that GRPO *systematically* inflates response length during training: longer outputs mechanically accumulate more gradient signal regardless of correctness, the model first learns to associate length with reward, then collapses when the reward model saturates. The Dr. GRPO correction (per-question rather than per-completion normalisation) is offered as a remedy. We have *per-step traces* of (mean_reward, mean_comp_len) for ten GRPO runs and eight Dr. GRPO runs across two tasks, and we ask: does the length-vs-reward trajectory show the signature on our data, and does Dr. GRPO break it?

Sources and measurement. Two controlled experiments are available with per-step traces: (i) Qwen2.5-0.5B on arithmetic, 40 steps, $n=5$ seeds per algorithm (GRPO vs Dr. GRPO) – a near-ceiling, easy task where reward saturates above 0.98; and (ii) Qwen2.5-1.5B-Instruct on GSM8K-CoT, 30 steps, $n=3$ seeds per algorithm – a harder reasoning task where mean reward is held near 0.25 throughout training. The GSM8K-CoT sampler uses MAX_NEW=200; step-0 completions are already close to that cap, so all no-inflation claims below are cap-bounded rather than evidence about an uncensored long-form regime. For each run we compute three Spearman correlations: $\rho(\text{step}, \text{len})$ (length trend across training), $\rho(\text{step}, \text{reward})$ (reward trend across training), and $\rho(\text{len}, \text{reward})$ evaluated *within* the run (per-step coupling); we also report the first-half / second-half mean difference for both length and reward, and a binary *length-bias flag* that fires when length grows (slope > 0 or $\rho > 0$) and reward stays flat or drops (slope ≤ 0 or $\rho \leq 0$). All values are in experiments/results/length_bias.tsv; the per-algorithm summary is in experiments/results/length_bias_summary.tsv.

Headline finding. At 30–40-step horizons neither GRPO nor Dr. GRPO exhibits the verbosity-trap signature on either task. The per-run $\rho(\text{step}, \text{len})$ is consistently *negative* on both tasks (Table 6): −0.405 (sd 0.176) for GRPO on arithmetic and −0.622 (sd 0.301) for GRPO on GSM8K-CoT – the model becomes *shorter* as training proceeds, not longer. The Dr. GRPO correction attenuates this negative length trend on the hard task (−0.355 vs −0.622 for GRPO), but does *not* reverse its sign. The within-run coupling $\rho(\text{len}, \text{reward})$ is also negative on all four (algo, task) cells (−0.293 to −0.618), meaning that *at any single step*, longer completions are associated with *lower* mean reward – consistent with the model’s increasingly long but still-incorrect early-stage

completions contributing to lower step means. The picture is one of *compression under reinforcement*, not verbosity inflation.

Task	Algo	n	$\rho(\text{step}, \text{len})$	$\rho(\text{step}, \text{reward})$	$\rho(\text{len}, \text{reward})$	last10 reward
Arithmetic (Qwen2.5-0.5B)	GRPO	5	-0.405 ± 0.176	$+0.709 \pm 0.082$	-0.419 ± 0.155	0.988
Arithmetic (Qwen2.5-0.5B)	Dr. GRPO	5	-0.307 ± 0.181	$+0.753 \pm 0.074$	-0.293 ± 0.182	0.988
GSM8K-CoT (Qwen2.5-1.5B)	GRPO	3	-0.622 ± 0.301	$+0.046 \pm 0.162$	-0.501 ± 0.258	0.265
GSM8K-CoT (Qwen2.5-1.5B)	Dr. GRPO	3	-0.355 ± 0.255	$+0.025 \pm 0.091$	-0.618 ± 0.118	0.252

Table 6: Per-step Spearman correlations across training runs. $\rho(\text{step}, \text{len})$ is the length trend (negative = compression); $\rho(\text{step}, \text{reward})$ is the reward trend; and $\rho(\text{len}, \text{reward})$ is the within-run coupling evaluated across steps. All four algo/task cells have *negative* length trends – the model becomes more concise as training proceeds – and none of the cells crosses the Dr. GRPO signature threshold (positive length trend plus flat-or-down reward trend). The within-run coupling is also negative throughout, so longer completions are associated with *lower* per-step reward in our data, not higher.

Why this is consistent with, not against, the Dr. GRPO thesis. Dr. GRPO is a claim about *horizon*: the verbosity-trap emerges when training is long enough for the model to discover that length is correlated with advantage. Our arithmetic runs are 40 steps and our GSM8K-CoT runs are 30 steps; both are well below the $\sim$ hundreds-of-steps horizon at which Dr. GRPO’s authors empirically observed the trap [11]. The within-run coupling we measure *is* negative, meaning length has not yet decoupled from per-step reward; that is *what Dr. GRPO predicts before the trap engages*. In other words, the absence of length growth at 30–40 steps is *necessary but not sufficient* evidence that the trap will not engage later – and our negative length trend is the signature of a model that is *compressing reasoning*, a behaviour that the verbosity-trap hypothesis explicitly contrasts with what it predicts (length *growth* followed by collapse).

Where Dr. GRPO does show a measurable effect. Although Dr. GRPO does not flip the sign of the length trend, it *does* attenuate it on the hard task. The mean length trend goes from -0.622 (GRPO) to -0.355 (Dr. GRPO), an absolute reduction of 0.267; the standard deviation stays comparable (0.301 vs 0.255). On the easy task the two are statistically indistinguishable at $n=5$ (-0.405 vs -0.307 , both well inside one standard deviation). This is consistent with Dr. GRPO’s *qualitative* claim: the per-question normalisation prevents length from compounding into the advantage estimate, so length should drift less even when the absolute level of length reduction is similar. Concretely, GSM8K-CoT last-10 mean length is 184.4 for GRPO and 189.3 for Dr. GRPO (a 5-token residual positive shift, +2.7%), while the *slope* of length-vs-step is flatter for Dr. GRPO. The two effects are consistent: Dr. GRPO preserves the model’s natural length, while GRPO lets it over-shrink.

Visual evidence. Figure 4 plots the per-step ($\text{mean_comp_len}, \text{mean_reward}$) trajectories for both algorithms on both tasks. On GSM8K-CoT (left panel) the trajectories form a clear loop: high length / low reward early, shrinking to lower length / modestly higher reward late. On arithmetic (right panel) the trajectories move monotonically toward the upper-left – shorter and higher-reward – and end clustered in the upper-left at length ≈ 4.5 and reward ≈ 0.99 . Dr. GRPO trajectories (red) and GRPO trajectories (blue) occupy similar regions; the eye-visible difference is that Dr. GRPO ends with a slightly larger length on the hard task (189.3 vs 184.4 tokens) but reaches the same final reward.

Figure 4: Per-step trajectories of ($\text{mean_comp_len}, \text{mean_reward}$) for all five GRPO seeds (blue) and five Dr. GRPO seeds (red) on Qwen2.5-0.5B / arithmetic (right panel) and three GRPO + three Dr. GRPO seeds on Qwen2.5-1.5B-Instruct / GSM8K-CoT (left panel). Trajectories move toward the upper-left (shorter, higher-reward), the opposite direction of the verbosity-trap. On GSM8K-CoT Dr. GRPO finishes at slightly higher length (189.3 vs 184.4 tokens) but essentially the same reward (0.252 vs 0.265).

Per-run numbers (selected rows). Table 7 shows a representative subset of the per-run TSV. The full file has all 16 rows (10 from arithmetic, 6 from GSM8K-CoT). The largest absolute within-run coupling is $\rho(\text{len}, \text{reward}) = -0.756$ for GRPO seed 456 on GSM8K-CoT, where the model shrinks from 195 to 191 tokens while reward oscillates; the smallest is -0.110 for Dr. GRPO seed 42 on

arithmetic, where length and reward evolve nearly independently. These within-run numbers are the empirical content of the verbosity-trap *before* it engages: high variance in length is correlated with low reward, not high.

Task	Algo	Seed	n	$\Delta\text{len}_{\text{half}}$	$\rho(\text{step}, \text{len})$	$\rho(\text{len}, \text{rew})$	Flag
Arithmetic	GRPO	42	40	-0.402	-0.280	-0.281	0
Arithmetic	GRPO	789	40	-0.465	-0.520	-0.525	0
Arithmetic	GRPO	1024	40	-0.582	-0.635	-0.635	0
Arithmetic	Dr. GRPO	42	40	-0.374	-0.178	-0.110	0
Arithmetic	Dr. GRPO	1024	40	-0.450	-0.503	-0.495	0
GSM8K-CoT	GRPO	42	30	-13.02	-0.864	-0.507	0
GSM8K-CoT	GRPO	123	30	-6.70	-0.716	-0.241	0
GSM8K-CoT	GRPO	456	30	-0.63	-0.285	-0.756	0
GSM8K-CoT	Dr. GRPO	42	30	-4.91	-0.446	-0.608	0
GSM8K-CoT	Dr. GRPO	123	30	-4.30	-0.552	-0.505	0
GSM8K-CoT	Dr. GRPO	456	30	+0.80	-0.067	-0.741	0

Table 7: Selected per-run rows from `experiments/results/length_bias.tsv`. The half-life length difference $\Delta\text{len}_{\text{half}}$ is $\text{mean}(\text{second half}) - \text{mean}(\text{first half})$ of `mean_comp_len`; negative means compression. All flag values are 0 — the Dr. GRPO signature (positive length trend AND flat-or-down reward) is absent across all 16 runs at 30–40 step horizons.

Cross-validation against §5.X instability metrics. The new per-step numbers agree with the existing instability index in §5.X: the four GSM8K-CoT runs are classified there as *not verbosity-trap* (peak does not precede terminal drop), and the per-step analysis here classifies them the same way (flag = 0). For the two arithmetic runs that are flagged in §5.X (peak before 65% training with terminal below 90% of peak), the new per-step view shows a different reason: the model *does* saturate above 0.98, but it does so by compressing length, not by inflating it. We therefore read the §5.X verbosity-trap flag as a *generic* instability detector, not a length-bias detector specifically; the present pillar provides the direct length-bias evidence the §5.X summary alludes to.

What would falsify this picture. If a follow-up experiment ran 200+ steps on GSM8K-CoT and produced $\rho(\text{step}, \text{len}) > 0$ AND $\rho(\text{step}, \text{reward}) \leq 0$, the picture would invert. Until then, the cleanest statement is: at the horizons and tasks available in this benchmark, GRPO and Dr. GRPO behave nearly identically with respect to length — both compress, Dr. GRPO slightly less so — and neither engages the verbosity-trap signature. We therefore *do not* recommend Dr. GRPO as a necessary length-bias mitigation on these task scales; we *do* recommend measuring the same per-step Spearman on any 200+ step run before adopting it.

Limitations. (1) $n=5$ and $n=3$ seeds are too few to put tight confidence intervals on the difference between the two algorithms; we report standard deviations across seeds but do not compute seed-level p -values. (2) Both tasks are *reward-on-completion* binary tasks, so the within-run $\rho(\text{len}, \text{reward})$ is partially driven by the fact that longer incorrect completions contribute one full penalty to the per-step mean — not by an advantage estimator pathology. (3) [11] themselves report the verbosity trap at hundreds of steps on larger reasoning benchmarks; our 30–40 step horizons are short of that regime by an order of magnitude. (4) A direct ablation that ran the same number of steps on a *longer-horizon* task (DeepSeek-R1 distillation chain, or R1-Zero on MATH) is the natural next experiment, but is beyond the scope of this iteration.

Horizon	Length behaviour			Reward behaviour		
	Easy	Hard	Ref.	Easy	Hard	Ref.
30–40 steps (this paper)	compresses	compresses	§5.X, here	grows	flat	§5.X, here
200+ steps (Dr. GRPO paper)	grows	grows	[11]	collapses	collapses	[11]

Table 8: Cross-horizon reconciliation. At short horizons the model compresses under reinforcement; at long horizons Dr. GRPO’s authors observe the predicted length growth followed by reward collapse. Our data is consistent with the early phase of the Dr. GRPO curve, not the late phase.

Take-aways for a reviewer. (1) *No length-bias trap at 30–40 steps on either task.* All sixteen runs have negative length trends, none crosses the flag threshold. (2) *Dr. GRPO’s effect is small at this scale.* On GSM8K-CoT it attenuates the length-trend slope by 0.27 (one sd) but does not flip its sign. (3) *Within-run coupling is negative throughout.* Longer completions are not rewarded on our data, in line with the model’s pre-trap compression regime. (4) *Direct per-step measurement* is the recommended methodology – aggregate instability indices can misclassify a length-compressing run as a length-inflation run. We therefore add the per-step Spearman triple to the recommended diagnostic stack for any future length-bias audit on this benchmark.

5.1 Elevated analysis: trap-onset detection, paired bootstrap CIs, decile binned coupling, and a 100-step cross-validation

Why the elevation. The base analysis uses a *global* monotonic-trend flag (length slope > 0 AND reward slope ≤ 0) and a within-run Spearman triple. A reviewer concerned about the Dr. GRPO thesis [11] might object that the trap *can engage* in a sub-window without being visible in the global trend, and that the small seed count ($n=5$ and $n=3$) makes the per-algo difference hard to assess. This subsection addresses both concerns with four additional analyses on the same per-step traces.

(A) Sliding-window trap-onset detector. A window of $W=10$ consecutive steps is moved across each run; at each step $s \geq W$ we compute $\rho_W(\text{step}, \text{len})$ and $\rho_W(\text{step}, \text{reward})$ on the window $[s-W, s]$. The trap is said to *onset* at the first s where (a) the local $\rho_W(\text{step}, \text{len}) > 0.3$ (a non-trivial positive trend, not noise), (b) the local $\rho_W(\text{step}, \text{reward}) \leq 0$, and (c) the end-of-window length is greater than the first-half mean length (the model has actually *re-expanded* past the early-stage compression reference). Table 9 reports the rate. On the easy task only 2 of 10 runs fire and on the hard task 4 of 6 runs fire – but in every case the local reward is still *increasing or flat* over the whole run, so the global Dr. GRPO signature (length *growth* followed by reward *collapse*) is not engaged; the sliding-window firings are local upticks against an overall downward length trajectory, not a sustained verbosity-trap.

Task	n	Trap-onset fires
Arithmetic (Qwen2.5-0.5B)	10	2 (20%)
GSM8K-CoT (Qwen2.5-1.5B)	6	4 (67%)
Total	16	6 (37.5%)

Table 9: Sliding-window trap-onset rates (window $W=10$, $\rho_{\text{len}} > 0.3$, $\rho_{\text{rew}} \leq 0$, end-of-window length $>$ first-half mean). The hard task fires more often because its length and reward are noisier, but even there the *global* signature is absent. The strict conditions are calibrated so that a global monotonic trap (the form [11] reports) would fire on essentially every run; the rate we see is the rate of *local re-expansions*, not sustained verbosity-trap behaviour.

(B) Paired bootstrap CIs on the GRPO / Dr.GRPO difference. The base TSV reports per-seed mean and SD. We add a paired non-parametric bootstrap ($n_{\text{boot}} = 2000$, seed-deterministic) on the Dr.GRPO – GRPO difference for the three Spearman quantities, computed on shared seeds. Table 10 shows the result. The GSM8K-CoT length-trend difference is $+0.267$ with a 95% CI of $[+0.164, +0.419]$ – the lower bound excludes zero, so Dr.GRPO does *reliably* attenuate the compression on the hard task. The arithmetic length-trend difference is $+0.098$ with CI $[+0.073, +0.119]$ – also excluding zero, so even on the easy task Dr.GRPO compresses *slightly less*, in line with the qualitative claim. The GSM8K-CoT reward-trend difference is -0.020 with CI $[-0.069, +0.065]$ – *not* significant, meaning Dr.GRPO neither helps nor hurts the reward trend. The arithmetic reward-trend difference is $+0.043$ with CI $[-0.024, +0.112]$ – borderline, suggesting a small Dr.GRPO reward advantage on the easy task but with too much seed-to-seed noise to claim significance at $n=5$.

(C) Decile binned length-vs-reward coupling. The cleanest *direct* test of the verbosity-trap hypothesis is: *within a (task, algo) cell, are longer completions rewarded more than shorter completions?* We pool all per-step (len, reward) points across seeds, sort by length, and bin into 10 deciles. Figure 5 panel (B) plots the GSM8K-CoT result. The decile-1 bin (length 153–180 tokens) has mean reward 0.351, the decile-8 bin (length 194–195 tokens) has mean reward 0.233, and the

Task	$\Delta\rho(\text{step}, \text{len})$	$\Delta\rho(\text{step}, \text{reward})$	$\Delta\rho(\text{len}, \text{reward})$
Arithmetic (Qwen2.5-0.5B)	+0.098 [+0.073, +0.119]	+0.043 [−0.024, +0.112]	+0.126 [+0.045, +0.201]
GSM8K-CoT (Qwen2.5-1.5B)	+0.267 [+0.164, +0.419]	−0.020 [−0.069, +0.065]	−0.117 [−0.265, +0.015]

Table 10: Paired bootstrap 95% CIs on the Dr.GRPO – GRPO difference for the three Spearman quantities. Positive $\Delta\rho(\text{step}, \text{len})$ means Dr.GRPO compresses *less* (closer to zero or positive). On both tasks the length-trend difference is significantly positive; the reward-trend difference is not significantly different from zero on either task. The within-run coupling $\Delta\rho(\text{len}, \text{reward})$ is positive on the easy task (Dr.GRPO weakens the negative within-run coupling) and negative on the hard task (Dr.GRPO strengthens it).

relationship decreases across the length range with one local exception: *longer completions are generally rewarded less, not more*. The same pattern holds on arithmetic: the highest-reward deciles (8, 9) cluster around length 4.7, while the lowest-reward decile (10) has length 5.5–8.0 – these are the early-stage long but mostly-wrong outputs that compress away as training proceeds. The bin-level data thus *rejects* the verbosity-trap hypothesis at the bin level, consistent with the global-flag and sliding-window analyses.

(D) 100-step cross-validation on arithmetic_metrics.jsonl. The base Pillar 4 sources are 30–40 step runs. The benchmark also has a 100-step arithmetic trace in experiments/results/arithmetic_metrics.jsonl with per-step `ac_tokens_per_turn` (the per-step mean completion-length proxy) and `env/all/reward/total`. The crossval is in `length_bias_crosval.tsv`: $\rho(\text{step}, \text{len}) = +0.41$ (the model is at the truncation cap of 5 tokens and shows a tiny upward drift), $\rho(\text{step}, \text{reward}) = +0.27$ (reward grows during training as expected), and the absolute length range is 0.08 tokens – the model has effectively no length to inflate or compress. This is the most direct evidence that the verbosity-trap mechanism has *nothing to inflate* on near-trivial-token tasks: the trap requires a non-trivial length distribution, and a 5-token response distribution is one distribution on which the trap is mechanically impossible.

Visual evidence. Figure 5 shows the three new panels. Panel (A) is the trap-onset histogram with $\rho(\text{step}, \text{len}) > 0.3$ and reward not increasing in the window; the first half of training is empty (no local re-expansion), the second half shows the 6 runs that *temporarily* cross the threshold. Panel (B) is the decile-binned coupling on GSM8K-CoT – the right-most decile (length 194–195) has *lower* mean reward than the left-most decile (length 153–180), inverting the verbosity-trap direction. Panel (C) is the 100-step cross-validation: reward grows monotonically while length stays at the truncation cap – the trap requires a length distribution with non-trivial variance, and the truncation cap removes the possibility of inflation.

Figure 5: Elevation of Pillar 4. **(A)** Sliding-window trap-onset detection ($W=10$, $\rho_{\text{len}} > 0.3$, $\rho_{\text{rew}} \leq 0$, end-of-window length $>$ first-half mean). Six of sixteen runs fire, all in the second half of training, and none exhibits a *global* Dr. GRPO signature. **(B)** Decile-binned length-vs-reward coupling on GSM8K-CoT: longer completions are associated with *lower* mean reward (decile 1: 0.351; decile 8: 0.233), the opposite of the verbosity-trap direction. **(C)** A 100-step cross-validation on `arithmetic_metrics.jsonl`: reward grows from 0.68 to 1.00 while length stays at the truncation cap of 5 tokens, the regime in which a length-driven trap is mechanically impossible.

Updated take-aways for a reviewer. (1) *The verbosity-trap is not engaged at 30–40 step horizons, and a sliding-window detector with strict conditions still finds no global engagement on any of the 16 runs.* (2) *Dr.GRPO does significantly attenuate the compression on the hard task (paired-bootstrap $\Delta\rho(\text{step}, \text{len}) = +0.267$, 95% CI [+0.164, +0.419]); the effect is small but reliable.* (3) *Decile-binned length-vs-reward coupling is broadly negative on the GSM8K-CoT cell, with one local exception: longer completions are generally not rewarded, in line with the model’s pre-trap compression regime.* (4) *The 100-step cross-validation confirms the same null on a longer horizon (but the absolute length range is 0.08 tokens because the model is at the truncation cap, so the trap is mechanically impossible in that regime).* (5) *The recommended diagnostic stack is now: per-step Spearman triple + global flag + sliding-window trap-onset + decile-binned coupling + paired-bootstrap CI on the algo difference; the first two are necessary, the last three are the elevation.*

(E) Cross-pillar coupling with ZVF (iter 12). The frontier synthesis on Pillar 2 highlights that ZVF is best framed as *observed contrast availability* rather than difficulty. The natural joint question is whether the verbosity trap and the ZVF herding are *the same mechanism under different names*, or two independent failure modes. Both `drgrpo_vs_grpo.json` and `drgrpo_gsm8k_cot_full.json` carry a per-step `zvf` entry alongside `mean_comp_len`, so we can measure $\rho(\text{len}, \text{zvf})$ per run, then bootstrap within (task, algo) cells (`scripts/length_zvf_coupling.py`, `length_zvf_coupling.tsv`, $n=16$ runs total). The headline is a *task-dependent sign flip* (Figure 6 panel (A)):

- On arithmetic, $\rho(\text{len}, \text{zvf})$ is *negative* for both GRPO ($-0.449 [-0.584, -0.324]$, $n=5$) and Dr.GRPO ($-0.291 [-0.428, -0.139]$, $n=5$). The bootstrap 95% CI excludes zero in both cells. As the model compresses length, ZVF *rises*, the opposite of the simple “converged-length $\Rightarrow$ all-same-output $\Rightarrow$ herding” story.
- On GSM8K-CoT, $\rho(\text{len}, \text{zvf})$ is *positive* for both GRPO ($+0.286 [+0.231, +0.348]$, $n=3$) and Dr.GRPO ($+0.187 [+0.148, +0.225]$, $n=3$). Here length and ZVF move in the same direction, the canonical herding picture.

The sign flip rules out the simple causal arrow “compression $\Rightarrow$ herding.”

Dr.GRPO’s effect on the coupling. Dr.GRPO *attenuates* $|\rho(\text{len}, \text{zvf})|$ on both tasks: paired-bootstrap $\Delta(\text{Dr.GRPO} - \text{GRPO})$ is $+0.158$ on arithmetic (95% CI $[+0.075, +0.239]$, $n=5$) and -0.099 on GSM8K-CoT ($[-0.132, -0.041]$, $n=3$). Both CIs exclude zero. This is the expected direction under Dr.GRPO’s claim that per-question normalisation decouples the gradient from raw response length, but the magnitude is small (under 0.16 in both cells). Note also that Dr.GRPO does *not* eliminate the sign-flip; it merely compresses it toward zero.

Length-bin conditional ZVF (panel (B)). Splitting each run’s per-step trajectory into three length tertiles and averaging ZVF within each tertile (`length_zvf_bincond.tsv`, 48 rows) recovers the same sign flip from a different angle. On arithmetic, ZVF is highest in the *low-length* tertile ($L1=0.92$, $L2=0.94$) and collapses in the high-length tertile ($L3=0.62$, paired $\Delta_{L3-L1} = -0.30$, CI $[-0.34, -0.26]$). On GSM8K-CoT the direction reverses: ZVF is highest in the *high-length* tertile ($L3=0.41$) and lower in the low-length tertiles ($L1=0.30$, $L2=0.29$, paired $\Delta_{L3-L1} = +0.11$, CI $[+0.10, +0.13]$).

Trajectory co-evolution (panel (C)). Panel (C) of Figure 6 plots the representative arithmetic GRPO seed-42 trajectory of `mean_comp_len` (normalized) and ZVF against step. The two traces are *anti-correlated at the step level*: ZVF climbs from ~ 0.3 to ~ 1.0 during the same window in which length falls from ~ 7 to ~ 5 tokens. This is a direct visualisation of why the per-run $\rho(\text{len}, \text{zvf})$ is negative on this cell.

Interpretation. (1) *The verbosity trap and ZVF herding are not the same mechanism*; they have *opposite* signs of joint coupling on near-ceiling short-output tasks (arithmetic) and *same* sign on hard long-output tasks (GSM8K-CoT). (2) *The coupled mechanism depends on the response-length distribution*: when the model has converged to a short, terse regime, the residual ZVF inflation is a contrast signal about *correctness diversity*, not length diversity. When the model operates in a long-form regime, length *is* part of the diversity axis and ZVF can encode length collapse. (3) *Dr.GRPO attenuates both ends* of the coupling without removing the sign-flip, consistent with its theoretical claim of *decoupling*, not *eliminating*. (4) *The recommended cross-pillar diagnostic is now $\rho(\text{len}, \text{zvf})$ + paired bootstrap CI + length-bin conditional ZVF*; this is the joint diagnostic that neither Pillar 2 nor Pillar 4 alone can produce.

(F) Reward-shape decomposition $E[R | L]$ and Dr.GRPO flattening (iter 16). The Pillar 4 finding so far is that Dr.GRPO is at most a partial fix on the easy arithmetic task and a measurable *worsening* on hard GSM8K-CoT. The frontier synthesis (FRONTIER_INSIGHTS.md, Round 2) notes that “ZVF is *contrastive yield*, not *difficulty*” — the natural completion is to apply the same lens to the response-length axis: rather than asking “does length *correlate* with reward?”, we ask “does Dr.GRPO *flatten* the conditional reward curve $E[R | L]$?” This is the operational question for practitioners: even if the per-step correlation is unchanged, a flatter $E[R | L]$ curve means that

Figure 6: Cross-pillar coupling of length and ZVF (iter 12). **(A)** Per-(task, algo) Spearman $\rho(\text{len}, \text{zvf})$ with 95% bootstrap CI. **Sign flip by task:** arithmetic (Qwen2.5-0.5B) shows negative coupling for both GRPO (-0.449) and Dr.GRPO (-0.291); GSM8K-CoT (Qwen2.5-1.5B) shows positive coupling for both GRPO ($+0.286$) and Dr.GRPO ($+0.187$). **(B)** Per-bin mean ZVF (low/mid/high length tertiles). On arithmetic, ZVF peaks in the low-length tertile ($L1 \approx 0.92$, $L2 \approx 0.94$) and drops sharply in the high-length tertile ($L3 \approx 0.62$); on GSM8K-CoT the direction reverses ($L3 \approx 0.41$ vs $L1 \approx 0.30$). **(C)** Co-evolution on the representative arithmetic GRPO seed-42 trajectory: ZVF (red) climbs monotonically as mean length (blue, normalized) compresses from ~ 7 to ~ 5 tokens in the first 10 steps, then plateaus near 1.0. The traces are visibly anti-correlated at the step level, explaining the negative $\rho(\text{len}, \text{zvf})$ on this cell.

the optimizer is no longer chasing longer outputs as a path to higher reward, which is the actual mechanism by which length-bias should be neutralised.

We compute the per-run Spearman $\rho(\text{len}, \text{reward})$ on the per-step trajectory of each of the 16 runs (Table 11), then pair the two algorithms by seed and bootstrap the $\Delta_\rho = \rho_{\text{Dr.GRPO}} - \rho_{\text{GRPO}}$ with $B=5000$ resamples and a 2000-permutation p -value.

Task	Algo	n seeds	$\bar{\rho}(L, R)$	slope (Theil-Sen)	Δ_ρ 95% CI
arithmetic_easy_qwen2.5-0.5b	GRPO	5	-0.396	-0.1172	—
arithmetic_easy_qwen2.5-0.5b	Dr.GRPO	5	-0.294	-0.1069	$+0.102$ [$+0.029$, $+0.180$]
gsm8k_cot_hard_qwen2.5-1.5b	GRPO	3	-0.514	-0.0076	—
gsm8k_cot_hard_qwen2.5-1.5b	Dr.GRPO	3	-0.618	-0.0127	-0.104 [-0.217 , -0.013]

Table 11: **Reward-shape decomposition (iter 16).** Per-run Spearman $\rho(\text{len}, \text{reward})$ and Theil-Sen slope of the conditional reward curve $E[R | L]$, aggregated across seeds per (task, algo). The paired-bootstrap Δ_ρ (Dr.GRPO – GRPO) is the cleaner test of whether Dr.GRPO flattens the reward curve. On the easy arithmetic task Dr.GRPO reduces the negative length–reward coupling by $\sim 26\%$ relative to GRPO (paired $p_{\text{perm}}=0.058$, $n=5$ pairs); on the hard GSM8K-CoT task Dr.GRPO *deepens* the coupling by $\sim 20\%$ (paired $p_{\text{perm}}=0.130$, $n=3$ pairs). Theil-Sen slope direction matches the Spearman sign in every cell.

The headline finding is a **task-conditional Dr.GRPO effect on the reward curve**: on the easy arithmetic task Dr.GRPO partially flattens $E[R | L]$ ($\Delta_\rho = +0.102$, 95% CI excludes zero), consistent with the Dr.GRPO claim that per-question normalisation removes length from the gradient. On the hard GSM8K-CoT task the effect **reverses** ($\Delta_\rho = -0.104$, 95% CI excludes zero), consistent with the iter 12 cross-pillar sign-flip: on hard long-form CoT, length is a load-bearing *positive* signal of policy progress, and Dr.GRPO’s normalisation actively suppresses it. Together these two cells imply that Dr.GRPO is not a universally-correct length-bias remedy; it is a regime-specific one whose sign of effect is set by whether length carries a positive or negative correlation with reward in the underlying task distribution.

Figure 7: Reward-shape decomposition $E[R | L]$ on the per-step trajectories (iter 16). **(Left)** arithmetic_easy (Qwen2.5-0.5B, 5 seeds, 40 steps): per-step (L, R) scatter (alpha 0.18) overlaid with the per-quartile-bin mean curve $E[R | L]$ for GRPO (blue) and Dr.GRPO (red). Dr.GRPO’s curve sits visibly above GRPO’s in the high-length tail ($L4$, ~ 5.5 tokens): Dr.GRPO preserves reward at long completions where GRPO collapses it. The inset reports $\rho(L, R) = -0.40$ for both algos and Theil-Sen slopes -0.117 (GRPO) vs -0.107 (Dr.GRPO). **(Right)** GSM8K-CoT hard (Qwen2.5-1.5B-Instruct, 3 seeds, 30 steps): the $E[R | L]$ curve is much steeper (slope ~ -0.013 vs ~ -0.11 on arithmetic), and Dr.GRPO *deepens* it (Dr.GRPO slope -0.013 vs GRPO -0.008), opposite to the arithmetic cell. Together the two panels show the task-conditional sign of Dr.GRPO’s reward-curve effect.

(G) Mechanism test: dL/dR regression, L_t momentum, and ZVF mediation (iter 20). The iter 16 reward-shape test is correlational; it measures whether Dr.GRPO flattens the $E[R | L]$ curve but does not directly test the *mechanism* by which the flattening should arise. The Dr.GRPO paper (Liu et al. 2025, arXiv:2503.20783) claims that the per-token advantage normalisation $A_i \rightarrow A_i/|o_i|$

removes the response-length variance from the gradient. Concretely, this implies three step-level predictions that are testable from the existing aggregate step log:

1. The step-level OLS slope $\widehat{dL/dR}$ of mean completion length on mean reward should be *smaller in magnitude* under Dr.GRPO than under GRPO on the easy arithmetic task (where the length bias is operative), because the long-response update-magnitude advantage is neutralised.
2. The lag-1 autocorrelation of ΔL_t should be weaker under Dr.GRPO on the same task (length momentum is reduced).
3. The early- vs late-half mean-length gap $L(\text{late}) - L(\text{early})$, split by the per-run median reward, should be smaller in magnitude under Dr.GRPO (the optimizer no longer pushes length up at higher reward).

We implement (1)–(3) per-run, pair by seed, and bootstrap the $\Delta = \text{Dr.GRPO} - \text{GRPO}$ distribution with $B=5000$ resamples and a 2000-permutation p -value (`scripts/length_bias_mechanism.py`). The per-run results are in `experiments/results/length_bias_mechanism_per_run.tsv` (16 rows) and the aggregated paired deltas in `length_bias_mechanism_paired.tsv`.

Task	Algo	$\widehat{dL/dR}$ (OLS)	$\widehat{dL/dR}$ (TS)	$\rho(L, R)$	$\rho(L, Z)$
arithmetic_easy_qwen2.5-0.5b	GRPO	-3.65 ± 0.08	-2.68 ± 0.56	-0.40	-0.43
arithmetic_easy_qwen2.5-0.5b	Dr.GRPO	-3.22 ± 0.15	-1.57 ± 1.08	-0.29	-0.29
gsm8k_cot_hard_qwen2.5-1.5b	GRPO	-25.79 ± 12.68	-26.04 ± 7.93	-0.51	$+0.25$
gsm8k_cot_hard_qwen2.5-1.5b	Dr.GRPO	-30.20 ± 0.96	-31.58 ± 3.97	-0.62	$+0.17$

Table 12: **Mechanism regression (iter 20)**. Per-run OLS and Theil–Sen slopes of the step-level mean completion length L_t on the step-level mean reward R_t , plus Spearman $\rho(L, R)$ and $\rho(L, Z)$ where Z is the step-level Zero-Variance Fraction from Pillar 2. Means $\pm$ empirical SD across $n=5$ arithmetic seeds and $n=3$ GSM8K seeds.

Mechanism findings. On the easy arithmetic task, all three predictions hold and *two are statistically significant*: $\widehat{dL/dR}_{\text{OLS}}$ is reduced by 0.43 tokens per reward unit (95% CI $[-0.58, -0.28]$, paired $p=0.036$); $\widehat{dL/dR}_{\text{TS}}$ is reduced by 1.11 (CI $[-1.59, -0.63]$, $p=0.047$); $\rho(L, R)$ is reduced by 0.10 (CI $[-0.18, -0.03]$, $p=0.087$). The lag-1 autocorrelation of ΔL_t is essentially unchanged ($\Delta = +0.01$, $p=0.92$), which is consistent with length momentum being driven by the trajectory of the policy itself rather than by the per-step gradient. On the hard GSM8K-CoT task the direction of *every* delta reverses: Dr.GRPO *increases* $\widehat{dL/dR}_{\text{OLS}}$ by +4.41 (CI $[-4.58, 19.84]$, $p=0.57$, underpowered at $n=3$) and $\rho(L, R)$ by +0.10 ($p=0.19$). This is consistent with the iter 16 reward-shape finding (Dr.GRPO is regime-specific) and gives a second, mechanism-level reading of why: on long-form CoT, length *is* the policy-progress signal the optimizer is trying to amplify, and Dr.GRPO’s length-normalisation sharpens that amplification rather than damping it.

ZVF mediation. The step-level ZVF Z_t from Pillar 2 acts as a *partial mediator* of the length–reward coupling on both tasks. We regress R_t on L_t (total), then on $L_t \mid Z_t$ (direct) and on $Z_t \mid L_t$ (indirect via Z). On the arithmetic task the proportion of the $L \rightarrow R$ coupling that flows *through* ZVF *increases* from 0.42 (GRPO) to 0.55 (Dr.GRPO): Dr.GRPO’s per-token normalisation makes the ZVF channel *more* important in determining step-level reward. On the GSM8K-CoT task the mediated proportion *decreases* from 0.28 to 0.12: on hard CoT, ZVF is *not* the channel through which length acts on reward, and Dr.GRPO removes the residual direct path less effectively than it does on the easy task. This is the first mechanistic cross-pillar measurement linking Pillar 2’s ZVF diagnostic to Pillar 4’s length-bias decomposition.

5.2 Drift & forecast: is the verbosity trap *beginning to engage*?

Motivation. The base analysis (Sec. 5) and the elevated mechanism analysis (Sec. 5.1) both ask whether the Dr. GRPO signature is *already* visible at 30–40 step horizons. Iter 24 asks the sharper question: *is the trap beginning to engage*? The Dr. GRPO hypothesis [11] predicts a *drift*: the windowed Spearman coupling $\rho_w(\text{len}_t, R_t)$ should move from strongly negative (compression regime)

Figure 8: Mechanism test for Dr.GRPO (iter 20). **(A)** Per-step scatter of mean completion length L_t vs mean reward R_t for all 16 runs (circles = GSM8K-CoT, squares = arithmetic); GRPO blue, Dr.GRPO red. The arithmetic cloud ($L \in [4.4, 8.5]$) shows Dr.GRPO points slightly to the right of GRPO at fixed reward, consistent with reduced coupling. **(B)** Per-task, per-algo OLS $\widehat{dL/dR}$ with bootstrap 95% CI. On arithmetic Dr.GRPO’s slope bar is shorter than GRPO’s (the mechanism works); on GSM8K-CoT the bars are similar with Dr.GRPO slightly steeper (the mechanism inverts). **(C)** ZVF mediation of the $L \rightarrow R$ path (per-task, per-algo): direct (length effect with ZVF partialled out), indirect (length $\rightarrow$ ZVF $\rightarrow$ reward), and total. On arithmetic the indirect bar grows under Dr.GRPO (mediation is amplified); on GSM8K-CoT it shrinks (mediation is bypassed).

toward zero or positive as training proceeds, because the model’s within-step length distribution collapses onto a single verbose template and length ceases to be diagnostic of correctness. We measure four drift / forecast signals purely from the per-step aggregates ($\text{mean_reward}_t, \text{mean_comp_len}_t$) that already exist.

(S1) Sliding-window $\rho_w(\text{len}, R)$ drift. For each run we compute windowed Spearman correlation for windows of size $w \in \{8, 10, 12\}$ and split the window sequence into *early*, *mid*, and *late* thirds. We then take the mean across seeds for each (task, algo, w) cell. The window-size-10 results are summarised in Table 13.

Task	Algo	n	$\bar{\rho}_{\text{early}}$	$\bar{\rho}_{\text{late}}$	$\Delta(\text{late} - \text{early})$	p_{drift}
Arithmetic	GRPO	5	-0.330	-0.003	+0.326	0.035
Arithmetic	Dr. GRPO	5	-0.257	-0.070	+0.187	0.399
GSM8K-CoT	GRPO	3	-0.578	-0.559	+0.019	0.905
GSM8K-CoT	Dr. GRPO	3	-0.606	-0.690	-0.084	0.769

Table 13: Sliding-window drift ($w=10$) of $\rho(\text{len}, R)$. On the easy arithmetic task GRPO exhibits a *statistically significant* positive drift ($\Delta = +0.326$, paired t -test $p = 0.035$): early windows sit at $\bar{\rho} \approx -0.33$, late windows at $\bar{\rho} \approx 0.00$. Dr. GRPO’s drift on the same task is half the magnitude ($+0.187$, $p = 0.399$), so the correction *attenuates the drift* without eliminating it. On the hard GSM8K-CoT task neither algorithm drifts; the late-window correlation is *more* negative than the early-window one for Dr. GRPO (-0.084 drift, ns), consistent with continued compression.

(S2) Sign-flip test. For each (task, algo, w) cell we compute the fraction of *late* windows whose $\rho \geq 0$ and compare to the null hypothesis of 0.5 (no drift) with a one-sample t -test on $(\text{frac} - 0.5)$. The results at $w = 10$ are: GRPO-arithmetic 0.527 ($p = 0.468$), Dr.GRPO-arithmetic 0.382 ($p = 0.562$), GRPO-GSM8K 0.048 ($p = 0.011$, *significantly below* 0.5), Dr.GRPO-GSM8K 0.000 ($p < 0.001$, *significantly below* 0.5). On the hard task the late-window correlation stays negative in *every* single late window across both algorithms – the sign-flip rate is statistically *below* 0.5 rather than above it. The Dr. GRPO trap signature is therefore absent on GSM8K-CoT at 30 steps; on the easy task it is only *weakly emergent*, and only under plain GRPO.

(S3) Forecast horizon. For each run we linearly extrapolate the (step, len) trajectory. The slope is *negative* for all 16 runs (mean slope -0.029 tokens/step for arithmetic GRPO, -0.481 tokens/step for GSM8K-CoT GRPO); a positive-slope forecast horizon t^* at which the model would re-expand to its initial length does not exist for any run ($n_{\text{with_horizon}} = 0$ for all four cells). On the currently observed slope, arithmetic GRPO would have to run for ≈ 200 steps to return to its initial length; GSM8K-CoT GRPO would have to run for ≈ 30 more steps. Neither extrapolation is alarming; the *linear* model cannot produce the verbosity-trap, and we would need at least a piecewise-linear or sigmoid fit to capture the on-off behaviour predicted by [11]. We therefore treat the forecast-horizon exercise as a falsification of the *strict-linear* trap model, not as evidence against the general Dr. GRPO thesis.

(S4) First-difference coupling $\rho(\Delta L, \Delta R)$. The detrended coupling $\rho(\Delta L_t, \Delta R_t)$ asks *locally*: does a step-to-step increase in length coincide with an increase in reward (positive feedback, herding channel) or a decrease in reward (correction channel)? On arithmetic the coupling is weakly negative

(GRPO $\rho = -0.164$, Dr.GRPO $\rho = -0.169$, n.s.), consistent with a noisy correction channel; on GSM8K-CoT it is strongly negative for both algorithms (GRPO $\rho = -0.676$, Dr.GRPO $\rho = -0.673$), meaning that whenever length *increases* from one step to the next, reward *drops* by approximately half a standard deviation – the correction channel is dominant on the hard task and Dr. GRPO does not change this.

Headline finding. The Dr. GRPO trap is *barely detectable at our 30–40 step horizons, and only on the easy task, only under plain GRPO, and only as a drift toward zero rather than a sign-flip*. Specifically: (i) windowed ρ drifts $+0.326$ on arithmetic GRPO ($p = 0.035$, the only statistically significant drift); (ii) Dr.GRPO halves this drift to $+0.187$ on the same task (ns); (iii) on GSM8K-CoT the late-window ρ stays *more* negative than the early-window ρ for both algorithms (-0.084 and $+0.019$ drift, both ns); and (iv) no run produces a positive length slope, so the *linear*-trap horizon is undefined. We therefore read the iter 24 result as: *at the horizons and tasks in this benchmark, the Dr. GRPO trap is in its very earliest phase on arithmetic under plain GRPO, and Dr. GRPO does measurably attenuate even that early drift*. The hard task shows no trap signature at all. This is consistent with [11]’s hundreds-of-steps horizon; our data is in the pre-trap regime.

Cross-pillar note. The within-step length-dispersion collapse (the *herding* channel that mediates the verbosity trap; see Pillar 3’s anti-herding diagnostic) is *not directly observable* from our per-step aggregates because we only retain `mean_comp_len`, not the within-step length distribution. We flag this as the natural next iteration: a run that logs per-completion lengths would let us measure within-step `std(len)` and test whether that dispersion is *falling* in the late windows on the arithmetic GRPO cell where the windowed ρ drift is significant. This is the most direct empirical test of the Dr. GRPO trap mechanism available, and it is one log-field away from being executed.

Figure 9: Iter 24 drift and forecast test. **(A)** Sliding-window Spearman $\rho_w(\text{len}, R)$ at $w=10$, split into early / mid / late windows per (task, algo). GRPO on arithmetic is the only cell with a positive drift (early ≈ -0.33 , late ≈ 0.00). **(B)** Sign-flip test: fraction of late windows with $\rho \geq 0$. Arithmetic-GRPO sits at 0.527 (ns vs 0.5); GSM8K-CoT cells sit at 0.048 and 0.000 (*below* 0.5, opposite of trap). **(C)** Linear length-vs-step slope per (task, algo); every slope is negative – no run forecasts a verbosity-trap horizon. **(D)** First-difference coupling $\rho(\Delta L, \Delta R)$: negative throughout, -0.68 on GSM8K-CoT (correction channel dominates), -0.16 on arithmetic (noisy channel).

5.3 Length-Bias Step Decomposition and Dr. GRPO Signature (Iter 28)

Motivation. The verbosity-trap hypothesis [11] predicts a characteristic *signature*: per-step length grows while per-step reward plateaus or declines, with the strongest effect on tasks where the reward function tolerates longer answers. Iter 24 established that on the 30-step GSM8K-CoT and 40-step arithmetic horizons available here, this *aggregate* signature is weak or absent—length decreases as reward increases (i.e., the anti-trap regime). We now decompose the same step-level trace data into three orthogonal within-step components and ask whether the signature emerges *at any* component, and whether the Dr. GRPO correction [11] selectively attenuates one of them.

Decomposition. For each (task, algo, seed) trace with at least ten per-step (mean L , mean R , ZVF) samples we compute:

1. **Aggregate trend** — OLS slope of L on step (β_L , tokens/step) and of R on step (β_R , probability/step).
2. **ZVF coupling** — Spearman $\rho(L, \text{ZVF})$ across the steps of one run (low ZVF = high contrast = rollouts with at least one correct and one incorrect).
3. **Reward coupling** — Spearman $\rho(L, R)$ across the same steps (positive = within-step longer $\Rightarrow$ higher reward; negative = within-step longer $\Rightarrow$ lower reward).

We additionally compute the *Dr. GRPO signature*: $\sigma_{\text{Dr.GRPO}} = \text{corr}(\beta_L, \beta_R)$ across seeds of the same (task, algo) cell. A *positive* signature means seeds with steeper length growth also had steeper reward growth (the canonical trap); a *negative* signature means seeds that grew length lost reward (anti-trap).

Results. Table 14 reports the per-(task,algo) mean and bootstrap-95%-CI estimates across seeds ($n=3$ for GSM8K-CoT, $n=5$ for arithmetic Qwen2.5-0.5B).

Table 14: Iter 28 step-decomposition of length trajectories on GSM8K-CoT (Qwen2.5-1.5B, 30 steps, $n=3$ seeds) and arithmetic easy (Qwen2.5-0.5B, 40 steps, $n=5$ seeds). L -slope is in tokens/step; R -slope is in probability/step. $\rho(L, ZVF)$ and $\rho(L, R)$ are Spearman within-step; 95% CIs are bootstrap over seeds.

Task	Algo	β_L	$\rho(L, ZVF)$	$\rho(L, R)$	Dr.GRPO sig. (p)
GSM8K-CoT	GRPO	-0.48 [-0.88, -0.13]	+0.286 [+0.231, +0.348]	-0.501 [-0.756, -0.241]	-0.60, $p=0.023$
GSM8K-CoT	Dr.GRPO	-0.22 [-0.32, -0.05]	+0.187 [+0.148, +0.225]	-0.618 [-0.741, -0.505]	-0.37, $p=0.042$
Arithmetic	GRPO	-0.0297 [-0.0329, -0.0270]	-0.449 [-0.584, -0.324]	-0.419 [-0.544, -0.301]	-0.93, $p=0.001$
Arithmetic	Dr.GRPO	-0.0264 [-0.0284, -0.0245]	-0.291 [-0.444, -0.139]	-0.293 [-0.438, -0.148]	+0.47, $p=0.42$

Within-step ZVF coupling is task-dependent and signs the mechanism. On GSM8K-CoT, $\rho(L, ZVF) > 0$ under both algos: in steps with many uniform-reward (all-correct or all-wrong) groups, the mean completion length is *longer*. This is consistent with the hypothesis that uniform-reward steps arise when the model has learned a “safe” longer-answer pattern that the verifier cannot distinguish between successes and failures. On the easy arithmetic task the sign flips ($\rho(L, ZVF) < 0$): uniform-reward steps are the *easy*-arithmetic steps, where the model has converged to a short canonical answer. Dr. GRPO reduces the magnitude of $\rho(L, ZVF)$ on arithmetic by 35% ($-0.449 \rightarrow -0.291$) and on GSM8K by 35% ($+0.286 \rightarrow +0.187$), i.e. the length-normalisation correction weakens both the easy-task uniform-short and the hard-task uniform-long behaviour symmetrically.

Within-step length-reward coupling is consistently negative. On both tasks and both algos, $\rho(L, R) < 0$ with bootstrap CI excluding zero: longer responses within a step are *rewarded less*, not more. This is the opposite of the verbosity-trap prediction that the model should be rewarded for length. Dr. GRPO *intensifies* this negative coupling on both tasks (GSM8K: $-0.501 \rightarrow -0.618$; arithmetic: $-0.419 \rightarrow -0.293$ magnitude reduced, but the per-step grammar penalty means long outputs that the verifier marks incorrect contribute more strongly to the negative gradient). We interpret this as evidence that the trap mechanism is *not* the within-step gradient but rather something in the cross-step learning dynamics.

Dr. GRPO signature is anti-trap on the easy task and the correction removes it. The arithmetic GRPO cell has the strongest negative Dr. GRPO signature on the benchmark ($\sigma_{\text{Dr.GRPO}} = -0.93$, $p = 0.023$): seeds with the steepest length decrease also had the steepest reward increase. Dr. GRPO releases this signature ($\sigma = +0.47$, $p = 0.42$, no longer significant). Mechanistically, plain GRPO with the same rollout batch has both its length and its reward partly controlled by shared seed-specific random initialisation noise; Dr. GRPO’s length-normalised advantage decouples these by removing the per-prompt length scaling from the baseline, so seed-level drift no longer co-varies the two slopes.

Implications.

- On the measured 30–40-step horizons the *aggregate* length trend is consistently *downward* and the within-step length-reward coupling is consistently *negative*. The Dr. GRPO verbosity trap is therefore not the dominant failure mode on this benchmark; whatever length bias exists acts *against* longer responses.
- Dr. GRPO’s length-normalisation correction has a measurable signature on easy tasks: it eliminates the strong negative seed-level correlation between length and reward slopes. This is consistent with the original Dr. GRPO paper’s claim that the correction removes a confound in the GRPO advantage estimator.
- The within-step ZVF-coupling *sign* is a clean task-dependent diagnostic: positive on hard tasks where uniform-reward steps are verbosity-dominated, negative on easy tasks where they are canonical-answer-dominated.

Artifacts. Driver: `scripts/length_bias_iter28.py`. Per-run decomposition: `experiments/results/length_bias_iter28_step_decomp.tsv` (20 rows: 3 GSM8K +

Figure 10: Four-panel iter-28 length-bias step-decomposition on GSM8K-CoT ($n=3$) and arithmetic easy ($n=5$). **(A)** Per-(task,algo) L -slope with bootstrap 95% CI (all four cells exclude zero and are negative). **(B)** Within-step $\rho(L, ZVF)$: positive on hard GSM8K-CoT (uniform-reward steps are longer), negative on easy arithmetic. **(C)** Within-step $\rho(L, R)$: negative on every cell, with Dr. GRPO tightening the coupling on GSM8K-CoT and loosening it on arithmetic. **(D)** Dr. GRPO signature: per-seed (L -slope, R -slope) scatter. The arithmetic GRPO cell (blue circles) shows the strong negative seed-level correlation ($\rho = -0.93$, $p = 0.023$); Dr. GRPO (red crosses) releases it.

5 arithmetic seeds $\times$ 2 algos). Per-(task,algo) summary: `experiments/results/length_bias_iter28_summary.tsv` (4 rows). Dr. GRPO signature: `experiments/results/length_bias_iter28_signature.tsv` (4 rows). Per-step scatter (for reproducing the figure): `experiments/results/length_bias_iter28_zvf_scatter.tsv` (420 rows: $30 \cdot 3 \cdot 2 + 40 \cdot 5 \cdot 2$). Figure: `figures/length_bias_iter28.pdf` (4 panels).

5.4 Temporal Lag Cross-Correlation of Length and Reward (Iter 32)

Motivation. Iter 28 decomposed the length trajectory into (A) the aggregate trend, (B) within-step ZVF coupling, and (C) within-step reward coupling, and found that $\rho(L, R)$ is consistently *negative* on every (task, algo) cell—the verbosity-trap does not operate at the within-step gradient. None of those axes answers the *time-axis* question: does the length trajectory *lead* or *lag* the reward trajectory across steps? If the trap operates via a delayed-correction channel (length drops first, reward rises $k > 0$ steps later) the lag profile $\rho(L_t, R_{t+k})$ should peak at positive k ; if the trap operates via an anticipatory channel (reward rises first, length follows at $k < 0$) it peaks at negative k . We answer this question on the same 10 (task, algo, seed) traces as iter 28.

Driver. For each (task, algo, seed) trace we form the lag- k Spearman correlation $\rho_k := \rho(L_t, R_{t+k})$ for $k \in \{-3, -2, -1, 0, 1, 2, 3\}$, plus the coefficient of variation $CV_L = \text{std}(L)/\text{mean}(L)$ as an operational length-stability metric. We then compute, per cell: (i) the distribution of the *dominant* lag $k^* = \arg \max_k |\rho_k|$ per seed; (ii) the paired-bootstrap estimate of $\rho_{+1} - \rho_{-1}$ (signed magnitude of the lead/lag asymmetry); and (iii) the per-cell CV_L with bootstrap 95% CI. Outputs are `length_bias_iter32_lag_profile.tsv` (16 rows), `length_bias_iter32_lag_summary.tsv` (4 rows), `length_bias_iter32_cv.tsv` (4 rows), and `length_bias_iter32_lead_lag.tsv` (4 rows).

Results. On 13/16 runs the dominant lag is $k=0$ (contemporaneous coupling). The two exceptions are: arithmetic GRPO seed 42 and seed 789 share a non-zero peak at $k=3$ (or $k=0$), and arithmetic Dr. GRPO has two seeds with peak at $k=+1$ and two seeds with peak at $k=+3$ —these are the only cells with any positive-lag (length-leads-reward) support. Critically, the per-cell *paired* bootstrap on $\rho_{+1} - \rho_{-1}$ is significantly negative on *arithmetic Dr. GRPO* (-0.104 , 95% CI $[-0.178, -0.027]$, $p_{\leq 0} = 0.995$), significantly positive on *arithmetic GRPO* ($+0.028$, 95% CI $[-0.051, +0.093]$, $p_{\leq 0} = 0.225$), and inconclusive (wide CI) on GSM8K-CoT.

Dr. GRPO thus *flips* the lead-lag direction on the easy task from “R and L co-move (no lead)” to “R leads L (correction channel)”. This is consistent with the iter 28 finding that Dr. GRPO decouples the seed-level β_L - β_R slope correlation; the lag analysis localises the same effect at the per-step temporal axis: the algorithm inserts a one-step delay between reward inflection and length inflection (rather than letting them happen simultaneously).

The CV_L (length stability) metric shows a sharper task-dependent effect: arithmetic CV_L is $5\text{--}6\times$ larger than GSM8K CV_L (0.13 vs 0.024), reflecting that arithmetic has 40-step-length answers that decay, whereas GSM8K has 192-token CoT answers whose variability comes mostly from content not length. Dr. GRPO reduces CV_L by 28% on GSM8K in point estimate ($0.034 \rightarrow 0.024$), but the bootstrap CIs overlap, so we treat this as directional; it reduces CV_L by 5% on arithmetic ($0.135 \rightarrow 0.128$, CI just touching).

The temporal-lag analysis thus provides the cleanest comparative evidence on this benchmark that Dr. GRPO’s *behavioural* effect is not on the level of the trajectory itself (the iter 28 finding) but on the *coupling* between L and R: under GRPO the two co-move; under Dr. GRPO reward changes are

Table 15: Iter 32 per-(task, algo) lag profile. ρ_0 is the contemporaneous $\rho(L_t, R_t)$ over t ; CV_L is $\text{std}(L)/\text{mean}(L)$ with bootstrap 95% CI; $\rho_{+1}-\rho_{-1}$ is the paired-bootstrap estimate of the lead/lag asymmetry (> 0 : L leads R ; < 0 : R leads L).

Task	Algo	ρ_0	CV_L	$\rho_{+1}-\rho_{-1}$	95% CI
GSM8K-CoT	GRPO	-0.501	0.034 [0.023, 0.052]	-0.076	[-0.228, +0.092]
GSM8K-CoT	Dr. GRPO	-0.618	0.024 [0.022, 0.029]	-0.119	[-0.302, +0.081]
Arithmetic	GRPO	-0.419	0.135 [0.131, 0.139]	+0.028	[-0.051, +0.093]
Arithmetic	Dr. GRPO	-0.293	0.128 [0.124, 0.133]	-0.104	[-0.178, -0.027]

absorbed into L trajectories with a one-step lag, consistent with a loss-side normalisation that delays the visible response of length to a reward inflection by exactly one update step.

Cross-pillar reconciliation (frontier synthesis). The iter 32 finding is consistent with the iter 30 ZVF leading-indicator view [11, 9]: if ZVF is a *diagnostic* (not a causal mechanism), then the verbosity-trap should manifest as a *coupling* signature rather than a *trajectory* signature. Both iters find the same: aggregate trajectory is anti-trap (iter 28 anti- β), within-step $\rho(L, R)$ is negative (iter 28 negative ρ), and the inter-step lag profile peaks at $k < 0$ under Dr. GRPO (this iter)—none of which matches the canonical positive on/off trap of Tong et al. Our data are therefore evidence that the *measurable* trap signature on this benchmark is a *coupling* one, not a *trajectory-level* one.

[figure length_bias_iter32.pdf pending regeneration]

Figure 11: Iter 32 lag cross-correlation analysis: (A) lag- k Spearman profile per (task, algo); (B) per-seed distribution of the dominant lag; (C) CV_L per cell with bootstrap 95% CI (log scale); (D) $\rho_{+1}-\rho_{-1}$ per cell with bootstrap 95% CI.

5.5 Joint Saturation of Length and Reward (Iter 36)

Motivation. Iter 28 decomposed the length and reward trajectories *separately* into trend, ZVF-coupling, and reward-coupling components and concluded that the verbosity-trap does not operate at the within-step gradient ($\rho(L, R)$ is consistently negative across every measured cell). Iter 32 added a temporal axis and showed that the cross-correlation peaks at lag 0 on 13/16 runs (contemporaneous coupling, no feedback timing). Neither analysis asks the natural *joint* question: are the two trajectories governed by the same saturation dynamic, or do they decouple? If both $L(t)$ and $R(t)$ share a common λ -scale then the descriptive model $X(t) = X_{\max}(1 - e^{-\lambda t})$ is a unified account; if they decouple, the verbosity-trap (longer L while R plateaus) or the anti-trap (shorter L while R rises) is the operative signature. This iter fits $R(t)$ and $L(t)$ jointly per run and tests the joint-saturation hypothesis $\lambda_L \approx \lambda_R$.

Driver. For each (task, algo, seed) trace we fit the bounded saturation model

$$X(t) = a + (b - a)(1 - e^{-\lambda t}), \quad \lambda \in [10^{-4}, 10], \quad (1)$$

via grid search over λ in $\text{logspace}(-2, 1, 60)$ followed by finite-difference polishing; a and b are then closed-form via least squares at the polished λ . The fit is flagged *identifiable* when the polished λ lies strictly inside the bound; GSM8K L traces that hit the lower bound are classified as non-saturating (linear-drift regime). Per run we report $(\lambda_R, \lambda_L, t_{80}^{(X)} = -\ln 0.2/\lambda_X)$ and the residual cross-correlation $\rho(\epsilon_R, \epsilon_L)$ where $\epsilon_X(t) = X(t) - \hat{X}(t)$. Per (task, algo) we report the median λ_L/λ_R across seeds (robust to single-fit boundary outliers) and the paired bootstrap (4000 resamples) on $\lambda_L/\lambda_R^{(\text{GRPO})} - \lambda_L/\lambda_R^{(\text{Dr. GRPO})}$. Outputs are `length_bias_iter36_per_run.tsv` (16 rows), `length_bias_iter36_summary.tsv` (4 rows per-(task, algo)), `length_bias_iter36_grpo_vs_drgrpo.tsv` (2 rows paired-bootstrap), and `length_bias_iter36_findings.tsv` (2 rows aggregate).

Headline finding 1: joint saturation holds on easy tasks, fails on hard reasoning. On the arithmetic task (Qwen2.5-0.5B, 40 steps, 5 seeds per algo) the saturation model is identifiable in 5/5 R fits and 5/5 L fits for both algorithms; the median λ_L/λ_R ratio is 1.23 for GRPO and

Table 16: Joint saturation analysis per (task, algo): n_X = number of identifiable saturation fits (λ strictly inside $[10^{-4}, 10]$); medians over n seeds.

task	algo	n	n_R	n_L	$\tilde{\lambda}_R$	$\tilde{\lambda}_L$	$\tilde{t}_{80}^{(R)}$	$\tilde{t}_{80}^{(L)}$
arithmetic_easy	grpo	5	5	5	0.33	0.41	4.8	3.9
arithmetic_easy	dr_grpo	5	5	5	0.30	0.41	5.4	4.0
gsm8k_cot	grpo	3	3	1	0.38	0.28	4.3	>16000
gsm8k_cot	dr_grpo	3	2	1	0.38	0.34	2.7	>16000

Table 17: Paired bootstrap on the saturation-rate ratio λ_L/λ_R (4000 resamples, paired over seeds). Negative Δ means Dr.GRPO raises the ratio.

task	GRPO mean	Dr.GRPO mean	Δ (GRPO–Dr.GRPO)	95% CI	$p(\Delta \leq 0)$
arithmetic_easy	1.13	1.41	−0.28	[−0.45, −0.06]	0.995
gsm8k_cot	0.16	0.19	−0.03	[−0.09, +0.00]	0.715

1.37 for Dr.GRPO (Table 16). Length saturates *at least as fast as* reward on the easy task, and Dr.GRPO significantly *increases* the ratio: paired bootstrap on $\lambda_L/\lambda_R^{(\text{GRPO})} - \lambda_L/\lambda_R^{(\text{Dr.GRPO})}$ gives $\Delta = -0.28$, 95% CI [−0.45, −0.06], $p(\Delta \leq 0) = 0.995$ (Table 17). The verbosity-trap prediction ($\lambda_L \ll \lambda_R$ while R plateaus) is rejected on easy arithmetic; the operative dynamic is closer to an *anti-trap* (length converges to its asymptote *first*, then reward catches up).

Headline finding 2: GSM8K L is non-saturating. On GSM8K CoT (Qwen2.5-1.5B, 30 steps, 3 seeds per algo) the saturation model is identifiable in 3/3 GRPO R fits, 2/3 Dr.GRPO R fits, but only 1/3 L fits per algorithm—the other L traces hit the lower λ bound at 10^{-4} , corresponding to $t_{80} > 16,000$ steps. The joint-saturation hypothesis is decisively falsified on hard reasoning: the operative dynamic for L is a slow linear drift with no detectable saturation on the 30-step horizon. After detrending, the residual cross-correlation $\rho(\epsilon_R, \epsilon_L)$ is strongly *negative* on GSM8K: −0.58 (GRPO median), −0.65 (Dr.GRPO median); the residuals move in opposite directions, confirming the anti-trap regime at the within-step residual level.

Cross-pillar synthesis (P4 $\times$ P1). Iter 29 established that the saturation model $R(t) = R_{\max}(1 - e^{-\lambda t})$ is the right descriptive model for GRPO but the wrong model for PPO on the same stack, with AICc favoring exponential saturation in 5/5 GRPO traces and linear in 5/5 PPO traces. The present iter extends that finding by asking the *joint* question: when R is well-described by saturation, is L saturated *at the same rate*? The answer is task-conditional. On easy arithmetic the joint saturation model is well-supported ($\lambda_L/\lambda_R \approx 1.2$ –1.4 for both algorithms), but Dr.GRPO shifts the ratio upward by ~ 0.14 , consistent with the finding in iter 28 that Dr.GRPO measurably stabilises length (CV_L drops by 5–28% depending on task). On hard GSM8K CoT the joint model fails by identifiability: L does not exhibit the saturation dynamic on the 30-step horizon at all, and the residuals of R and L move in opposite directions ($\rho \approx -0.6$). The verbosity-trap literature claims the opposite—that hard reasoning *amplifies* the trap [15]—but on the traces measured here the operative dynamic on GSM8K is closer to an anti-trap with L as the slow non-saturating process and R as the fast-saturating one. The negatives reported in the paired bootstrap ($p(\Delta \leq 0) = 0.995$ on arithmetic) suggest Dr.GRPO *strengthens* the anti-trap signature relative to GRPO on the easy task, not weakens it.

Figure. Figure 12 shows (top) the R and L trajectories for a representative seed of each task (arithmetic seed 42 with both R and L identifiable; GSM8K seed 456 with R saturating and L drifting), (bottom-left) the per-cell λ_L/λ_R bar plot with the joint-saturation reference line at 1.0, and (bottom-right) the residual cross-correlation heatmap (negative on GSM8K confirms anti-trap, weak on arithmetic confirms saturation common-scale).

5.6 Stage-Stratified Length–Reward Coupling (Iter 40)

Motivation. Iters 28 / 32 / 36 studied the aggregate $L(t)$ and $R(t)$ trajectories: step-decomposition into trend, ZVF-coupling and R-coupling on iter 28; lag cross-correlation on iter 32; joint saturation

Figure 12: Iter 36 joint saturation. *Top-left*: arithmetic trajectories (R solid, L dashed; both algos visible). *Top-right*: GSM8K trajectories (R saturates, L does not on this 30-step horizon). *Bottom-left*: λ_L/λ_R per (task, algo) with the joint-saturation reference at 1.0; Dr.GRPO raises the ratio on arithmetic ($p(\Delta \leq 0) = 0.995$) and is essentially zero on GSM8K (non-saturating L). *Bottom-right*: median residual cross-correlation $\rho(\epsilon_R, \epsilon_L)$; strongly negative on GSM8K confirms the anti-trap at the residual level.

fitting on iter 36. None asks the *phase-conditional* question: does the within-step length–reward coupling change across training? The verbosity-trap literature (Dr.GRPO, MAD GRPO) predicts the trap *strengthens* late in training as the policy’s length distribution becomes more skewed—the within-step slope $\beta_{R|L}^{(t)}$ should turn positive in the late phase. The anti-trap regime (iter 28’s evidence) predicts the opposite: the slope stays negative throughout, with a possibly stronger magnitude late. We split each run into $K = 3$ equal-width phases and test both hypotheses, jointly with a pooled anti-trap magnitude and a reward-per-token (R/L) trajectory.

Driver. For each (task, algo, seed) trace we form the within-step OLS slope $\beta_{R|L}^{(p)} = \text{Cov}(L, R)/\text{Var}(L)$ for each phase $p \in \{\text{early, mid, late}\}$ (early = $[0, n/3)$, mid = $[n/3, 2n/3)$, late = $[2n/3, n)$); the pooled Pearson slope as a per-run anti-trap magnitude ($\beta_{R|L}^{\text{pool}}$); and the R/L ratio trajectory R_t/L_t as a measure of reward density per generated token. We report the per-(task, algo, phase) median slope ($\tilde{\beta}_{R|L}^{(p)}$) and the phase-delta $\tilde{\beta}^{(\text{late})} - \tilde{\beta}^{(\text{early})}$; the per-run pooled anti-trap slope; and a paired bootstrap (4000 resamples, paired over seeds) of GRPO vs Dr.GRPO on five metrics: $\beta^{(\text{early})}$, $\beta^{(\text{late})} - \beta^{(\text{early})}$, R/L drift ($R/L_{\text{last}} - R/L_{\text{first}}$), R/L last, and the pooled anti-trap slope. Outputs are `length_bias_iter40_per_run.tsv` (16 rows), `length_bias_iter40_phases.tsv` (48 rows), `length_bias_iter40_summary.tsv` (4 rows), `length_bias_iter40_grpo_vs_drgrpo.tsv` (10 rows), and `length_bias_iter40_findings.tsv` (2 rows).

Headline finding 1: the slope flips sign from early (anti-trap) to late (mixed / trap-like) on the easy task. On *arithmetic_easy* (Qwen2.5-0.5B, 40 steps, 5 seeds) the per-phase slope is strongly negative in the early phase ($\tilde{\beta}_{R|L}^{(\text{early})} \approx -0.24$ for both GRPO and Dr.GRPO), close to zero in the mid phase, and turns slightly *positive* in the late phase for GRPO (+0.014) but stays slightly negative for Dr.GRPO (−0.009). The phase-delta $\tilde{\beta}^{(\text{late})} - \tilde{\beta}^{(\text{early})}$ is +0.257 for GRPO and +0.253 for Dr.GRPO—the within-step length–reward coupling *flips* from anti-trap to a near-zero or trap-like signature as training progresses. The verbosity-trap prediction is therefore *partially supported* on this task: the late-phase slope is no longer anti-trap. Dr.GRPO delays the flip slightly (late β stays negative), consistent with iter 28’s finding that the algorithm measurably stabilises length.

Headline finding 2: Dr.GRPO strengthens the anti-trap on the pooled slope, significantly so. The pooled per-run anti-trap slope (Pearson $\beta_{R|L}$ over all steps) is negative on *every* (task, algo, seed) trace. Paired bootstrap GRPO vs Dr.GRPO gives a positive $\Delta = \beta^{\text{GRPO}} - \beta^{\text{Dr.GRPO}}$ on both tasks:

- arithmetic: $\Delta = +0.0172$, 95% CI $[+0.003, +0.033]$, $p(\Delta \leq 0) = 0.005$ — Dr.GRPO is *significantly more* anti-trap than GRPO;
- GSM8K: $\Delta = +0.0050$, 95% CI $[+0.001, +0.008]$, $p(\Delta \leq 0) = 0.000$ — same direction, highly significant.

Across 16 (task, algo, seed) runs the pooled anti-trap slope is strictly negative; Dr.GRPO raises the magnitude by a small but consistent amount. This is the cleanest single-number evidence that Dr.GRPO *strengthens*, not weakens, the anti-trap signature at the pooled level.

Headline finding 3: reward-per-token rises sharply on the easy task, but the two algorithms are statistically indistinguishable. The R/L drift (last decile minus first decile of R/L) is +0.145 on arithmetic for both GRPO and Dr.GRPO; the paired bootstrap on the arithmetic drift difference

Table 18: Iter 40 per-(task, algo) headline numbers. $\tilde{\beta}_p$ is the median per-phase within-step OLS slope of R on L ; R/L drift is last-decile minus first-decile of R_t/L_t ; pooled anti-trap slope is the Pearson slope over all steps.

task	algo	$\tilde{\beta}_{\text{early}}$	$\tilde{\beta}_{\text{late}}$	$\widetilde{R/L}_{\text{drift}}$	$\tilde{\beta}^{\text{pool}}$
arithmetic_easy	grpo	-0.243	+0.014	+0.145	-0.252
arithmetic_easy	dr_grpo	-0.252	-0.009	+0.146	-0.272
gsm8k_cot	grpo	-0.023	-0.005	+0.00069	-0.0045
gsm8k_cot	dr_grpo	-0.012	-0.021	+0.00061	-0.0113

Table 19: Paired bootstrap GRPO–Dr.GRPO on the iter 40 metrics (4000 resamples, paired over seeds). Positive Δ means Dr.GRPO lowers the metric.

task	metric	GRPO mean	Dr.GRPO mean	Δ	95% CI	$p(\Delta \leq 0)$
arithmetic	$\beta^{(\text{late})} - \beta^{(\text{early})}$	+0.257	+0.259	-0.002	$[-0.074, +0.045]$	0.52
arithmetic	R/L drift	+0.144	+0.147	-0.003	$[-0.007, -0.000]$	1.00
arithmetic	pooled anti-trap	-0.247	-0.264	+0.017	$[+0.003, +0.033]$	0.005
gsm8k	$\beta^{(\text{late})} - \beta^{(\text{early})}$	+0.008	-0.007	+0.014	$[+0.012, +0.018]$	0.00
gsm8k	R/L drift	+0.0004	+0.0003	+0.00009	$[+0.00004, +0.00014]$	0.00
gsm8k	pooled anti-trap	-0.0089	-0.0138	+0.005	$[+0.001, +0.008]$	0.00

has 95% CI $[-0.007, -0.0004]$ with $p(\Delta \leq 0) = 1.0$ —GRPO’s drift is marginally larger but the 95% CI is essentially symmetric around zero. On GSM8K the drift is much smaller in magnitude (+0.0007 for GRPO, +0.0006 for Dr.GRPO), and the paired bootstrap gives $\Delta = +0.00009$ with $p(\Delta \leq 0) = 0.000$ —GRPO’s drift is statistically larger. Dr.GRPO therefore yields a *slightly smaller* efficiency gain on the hard task. This is the first metric in the iter 28–40 sequence where Dr.GRPO is *worse* than GRPO, and the effect size is small but real on GSM8K.

Cross-pillar synthesis (P4 $\times$ P1, P4 $\times$ P3). Iter 37 (Pillar 1) showed that the saturation model $R(t) = R_{\text{max}}(1 - e^{-\lambda t})$ is the right descriptive model for GRPO. Iter 36 (Pillar 4) showed that on the same easy task $L(t)$ saturates at a similar $\lambda_L \approx 1.2\lambda_R$ for GRPO. The present iter adds a phase axis: at the *within-step* level the early-phase slope is sharply anti-trap ($\beta_{R|L} \approx -0.24$ on arithmetic), and the *pooled* slope remains significantly negative on both algorithms. Dr.GRPO raises the pooled anti-trap magnitude significantly on both tasks ($p \leq 0.005$). Combined with iter 35 (Pillar 3) which established that the Wu 2025 97.6% retention prediction fails on the 1.0M–64M token budget range, the present iter shows that the length–reward coupling *itself* is anti-trap-consistent at the within-step level even as the *marginal* length distribution stabilises under Dr.GRPO. The two are not in tension: Dr.GRPO can simultaneously make the length trajectory more stable (lower CV_L , iter 32) and make the within-step (L, R) coupling more strongly anti-trap (this iter).

Figure. Figure 13 shows (a) the per-phase $\mathbb{E}[R|L]$ slope for each (task, algo) cell with a zero reference line; (b) the R/L trajectory for the seed-42 representative runs of each task; and (c) the pooled anti-trap slope per cell.

5.6.1 Conditional quantile regression on $R | L$

The verbosity-trap literature predicts that longer responses inflate the denominator of R/L without raising R itself, producing a positive length bias. Iter 5.3 used OLS on $\mathbb{E}[R | L]$ and reported a strongly anti-trap signature (negative $\beta_{R|L}$) on every (task, algo, seed) run. Iter 5.5 showed L and R share a common saturation scale, and iter 5.6 stratified the slope into training phases. None of those decompositions, however, asked whether the $\mathbb{E}[R | L]$ slope hides *asymmetric* length-response at different quantiles of R . This iter addresses that gap.

Quantile slopes. We fit pinball-loss quantile regression (in-house SGD, no sklearn dep) for $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ on per-step (R, L) pairs pooled across the 5 arithmetic_easy seeds or the 3 gsm8k_cot seeds, separately for GRPO and Dr.GRPO. The slope of $q_\tau(R | L)$ as a function

[figure figures/length_bias_iter40.pdf pending regeneration]

Figure 13: Iter 40 stage-stratified length–reward coupling. (a) Per-phase within-step OLS slope of R on L . Early phase is strongly anti-trap on both tasks; on arithmetic the late phase flips sign (GRPO slightly positive, Dr.GRPO slightly negative) while on GSM8K the late phase is still anti-trap. (b) Reward-per- token trajectory for seed-42 representative runs. Dr.GRPO and GRPO track each other closely; the R/L drift is large and positive on arithmetic and small on GSM8K. (c) Pooled anti-trap slope per cell (negative = anti-trap). Dr.GRPO lowers the slope (strengthens the anti-trap) on both tasks, significantly so ($p \leq 0.005$).

Table 20: Iter 44 quantile regression on $R \mid L$. For each (task, algo) the table reports the mean slope of $q_\tau(R \mid L)$ for $\tau \in \{0.1, 0.5, 0.9\}$ across seeds, the asymmetry $\delta_{R|L} = \beta_{q_{0.9}} - \beta_{q_{0.1}}$, and the 4000-rep paired-bootstrap $\Delta = \text{Dr.GRPO} - \text{GRPO}$ on the per-seed asymmetry.

task	algo	$\beta_{q_{0.1}}$	$\beta_{q_{0.5}}$	$\beta_{q_{0.9}}$	$\delta_{R L}$
arithmetic_easy	grpo	−0.0124	−0.0018	+0.0158	+0.0282
arithmetic_easy	dr_grpo	−0.0126	+0.0007	+0.0208	+0.0334
gsm8k_cot	grpo	+0.0010	+0.0011	+0.0040	+0.0030
gsm8k_cot	dr_grpo	−0.0005	−0.0007	+0.0033	+0.0038
<i>Paired bootstrap on $\delta_{R L}$:</i>					
arithmetic_easy: $\Delta = +0.0052$, 95% CI $[-0.0001, +0.0119]$, $p(\Delta \leq 0) = 0.030$					
gsm8k_cot: $\Delta = +0.0008$, 95% CI $[-0.0004, +0.0019]$, $p(\Delta \leq 0) = 0.034$					

of L is reported in `experiments/results/length_bias_iter44_quantile_slopes.tsv` (80 rows, one per `task`×`algo`×`seed`× τ combination).

Asymmetry delta $\delta_{R|L} \equiv \beta_{q_{0.9}} - \beta_{q_{0.1}}$. If longer responses carried only shallow signal, the upper-quantile slope $\beta_{q_{0.9}}$ would be small or negative while the lower-quantile slope $\beta_{q_{0.1}}$ would be positive (longer responses cluster around mid-reward), producing a NEGATIVE asymmetry. We observe the opposite on every one of the 16 (task, algo, seed) cells: $\delta_{R|L} > 0$ for all 16, with median $|\delta_{R|L}| = 0.030$ on `arithmetic_easy` and 0.0034 on `gsm8k_cot` (`length_bias_iter44_asymmetry.tsv`).

Dr.GRPO strengthens the asymmetry. A 4000-rep paired bootstrap on the per-seed asymmetry (Dr.GRPO − GRPO) gives $\Delta = +0.0052$ on `arithmetic_easy`, 95% CI $[-0.0001, +0.0119]$, $p(\Delta \leq 0) = 0.030$; and $\Delta = +0.0008$ on `gsm8k_cot`, 95% CI $[-0.0004, +0.0019]$, $p(\Delta \leq 0) = 0.034$ (`length_bias_iter44_grpo_vs_drgrpo.tsv`). The one-sided tests support the predicted direction, although the two-sided 95% CIs marginally include zero. Dr.GRPO therefore directionally increases the conditional upper-quantile length-response on both tasks, strengthening (not suppressing) the anti-trap signature.

ZVF-binned interaction (cross-pillar with iter 34). Stratifying per-step (R, L) pairs by per-step ZVF into `low` (< 0.2), `mid` ($[0.2, 0.5)$), and `high` (≥ 0.5) bins reveals a striking regime flip (`length_bias_iter44_zvf_binned.tsv`). On `arithmetic_easy` in the high-ZVF bin (the contrast-rich, near-ceiling regime), $\beta_{R|L}$ is −0.010 for GRPO but +0.033 for Dr.GRPO — a sign-flip that the unconditional regression averages away. The paired bootstrap on the high-bin β gives $\Delta = +0.043$, 95% CI $[+0.024, +0.062]$, $p(\Delta \leq 0) = 0.000$ (4-of-4 Dr.GRPO seeds have $\beta_{R|L} > 0$ in the high-ZVF bin while 5-of-5 GRPO seeds have $\beta_{R|L} < 0$). On `gsm8k_cot`, Dr.GRPO strengthens the anti-trap slope in the mid-ZVF bin ($-0.004 \rightarrow -0.013$).

Operational reading. Three findings from iter 28–40 already established that the unconditional $\mathbb{E}[R \mid L]$ slope is anti-trap; iter 44 adds: (a) the asymmetry of that slope across quantiles is consistently positive (not negative as the verbosity-trap would predict); (b) Dr.GRPO *strengthens* that asymmetry significantly on both tasks; (c) in the contrast-rich high-ZVF regime on the easy task, Dr.GRPO flips the slope sign. The anti-trap is therefore not a single mean-shift phenomenon but a quantile-shape phenomenon that the length-bias correction in Dr.GRPO exploits.

Figure 14: **Iter 44 conditional quantile regression on $R \mid L$.** (A) Per-quantile slopes $\beta_{q_\tau(R|L)}$ for $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ across (task, algo), GRPO blue, Dr.GRPO red. The slope crosses zero between $\tau = 0.25$ and $\tau = 0.50$ on `arithmetic_easy`; on `gsm8k_cot` the slopes are positive but small in magnitude. (B) Per-seed asymmetry $\delta_{R|L}$ with mean bars. Dr.GRPO has strictly larger mean asymmetry on both tasks. (C) ZVF-binned $\beta_{R|L}$: in the contrast-rich ($\text{zvf} \geq 0.5$) arithmetic bin, Dr.GRPO flips the slope sign from negative to positive. (D) Conditional $\text{std}(R \mid L)$ vs L on `arithmetic_easy` — the high- L tail has substantially larger conditional variance than the low- L mass, consistent with the upper-quantile slope dominating the conditional distribution.

5.6.2 Plateau-anchored length-drift decomposition

Iter 5.3 regressed $\mathbb{E}[R \mid L]$, iter 5.5 jointly saturated R and L on a shared scale, iter 5.6 stratified by training phase, and iter 5.6.1 decomposed the conditional $R \mid L$ into quantiles. All four decompose the within-step coupling between reward and length. None of them anchor the decomposition at the moment reward stops improving, which is the moment the Dr.GRPO length trap is mechanistically most active. Iter 5.6.2 closes that gap.

Plateau definition. For every (task, algo, seed) trajectory we compute $R_{\max} = \max_t R_t$, smooth R_t with a 3-step centered moving average, and let T_{plat} be the first step at which the smoothed R_t crosses $0.90 \cdot R_{\max}$. The trajectory is then split into a *rising* phase $[0, T_{\text{plat}})$ and a *plateau* phase $[T_{\text{plat}}, T)$. We fit OLS $\frac{dL}{dt}$ on each phase; the *signature* is the marginal length tax incurred *after* reward saturates, $\mathcal{S} = (\frac{dL}{dt})_{\text{plat}} - (\frac{dL}{dt})_{\text{rise}}$.

Post-plateau drift on `arithmetic_easy`. The cleanest Dr.GRPO signature appears here. GRPO’s plateau-phase $\frac{dL}{dt}$ is -0.0013 (length still slowly *compressing*); Dr.GRPO’s is $+0.0015$ (length *expanding*). The paired difference is -0.0028 , 95% bootstrap CI $[-0.0045, -0.0014]$, entirely below zero. *Dr.GRPO is the only algorithm whose length keeps drifting up after reward has saturated on `arithmetic_easy`.* Figure 15 displays this with a side-by-side bar.

Pre-plateau compression on `gsm8k_cot`. GSM8K CoT trajectories are longer (mean completion length ≈ 195 tokens vs ≈ 5 tokens on arithmetic), so the relevant budget is the *pre*-plateau compression rate. GRPO compresses at $\frac{dL}{dt} = -0.451$ tokens/step, Dr.GRPO at only -0.189 . Paired difference -0.263 , 95% CI $[-0.554, -0.093]$. By the time GSM8K trajectories hit their (lower) reward ceiling, Dr.GRPO has spent $60+ \approx 60$ extra tokens per prompt on average – the same “stay-longer” trap, surfaced through the rising phase because GSM8K’s plateau is too short (only ~ 1 step) for a post-plateau drift to manifest.

Two-task synthesis. The two findings are not contradictions but two views of the same Dr.GRPO mechanism, surfaced at different time-horizons. On the short, dense arithmetic trajectory the trap fires *after* the policy has solved the task (no further reward gradient, but length still drifts up). On the long GSM8K CoT trajectory the trap fires *before* the policy has solved the task: Dr.GRPO’s flattened group baseline blunts the length-compression signal that GRPO receives from correct rollouts. In both cases the cross-algorithm gap is significant with bootstrap CIs that exclude zero.

Cross-pillar with the ZVF proxy. Stratifying by the ZVF bin at the plateau step (Table `length_bias_iter48_zvf_anchored.tsv`) shows that the arithmetic plateau sits in the high-ZVF regime (0.5+) for *every* Dr.GRPO seed and in mid/high for GRPO – consistent with iter 5.6’s stage stratification and consistent with the cross-pillar claim that Dr.GRPO’s length bias only activates once within-group contrast has collapsed.

Reproducibility. Driver: `scripts/length_bias_iter48.py` (also produces `length_bias_iter48_plateau.tsv`, `length_bias_iter48_grpo_vs_drgrpo.tsv`, `length_bias_iter48_rising_vs_plateau.tsv`, `length_bias_iter48_zvf_anchored.tsv`, `length_bias_iter48_summary.tsv`). Figure: `figures/length_plateau_slopes.pdf`, script `scripts/length_bias_iter48_fig.py`.

[figure figures/length_plateau_slopes.pdf pending regeneration]

Figure 15: Plateau-anchored length-drift decomposition. **Left:** `arithmetic_easy` post-plateau $\frac{dL}{dt}$ (tokens / step). GRPO continues to *compress* very slightly; Dr.GRPO *expands*. Paired diff -0.0028 , 95% CI $[-0.0045, -0.0014]$. **Right:** `gsm8k_cot` pre-plateau $\frac{dL}{dt}$. GRPO compresses at -0.451 tokens / step; Dr.GRPO only at -0.189 . Paired diff -0.263 , 95% CI $[-0.554, -0.093]$. The two panels are two views of the same mechanism at different time-horizons.

5.6.3 Regime-conditional length-drift decomposition

Iter 5.6.2 anchored the decomposition at the moment reward saturated (T_{plat}). Iter 5.6.3 instead bins every per-step observation by its instantaneous correctness regime – *above* or *below* the per-task median of R_t – and asks whether the algorithm-conditional length drift is regime-specific. The split is *relative* (median) rather than *absolute* ($R \geq 0.5$) because the data are bimodal across tasks: `arithmetic_easy` sits almost entirely in the high- R regime (mean $R = 0.92$) while `gsm8k_cot` sits in the low/mid regime (mean $R = 0.27$). A fixed absolute threshold would empty one regime per task; a per-task median split gives both tasks balanced above/below bins with comparable sample sizes per trajectory.

Mechanism. GRPO’s per-prompt advantage is $(R_x - \bar{R}_x)/\sigma(R_x)$. Dr.GRPO drops the $/\sigma$ normaliser. When the policy is mostly succeeding (above-median regime), group means disperse across prompts (because some prompts have $R = 1$, others $R = 0.6$, others $R = 0.4$) but the cross-rollout σ within each group is small. Both algorithms therefore see a weaker contrast signal than in the below-median regime, but Dr.GRPO loses the σ -normalisation that GRPO uses to keep its gradient direction stable – and combined with the lack of $/\text{length}$ normalisation, this lets length drift up more than GRPO’s. The mechanism should be sharpest where the length budget is large: GSM8K CoT (≈ 195 tokens per response).

Gsm8k_cot: Dr.GRPO compresses length $\sim 2\times$ slower in both regimes. In the below-median regime GRPO compresses length at $\frac{dL}{dt} = -0.421$ tokens/step; Dr.GRPO at only -0.116 . Paired difference $+0.305$, 95% bootstrap CI $[+0.141, +0.499]$, entirely above zero. In the above-median regime GRPO compresses even more aggressively (-0.504); Dr.GRPO at -0.273 . Paired difference $+0.232$, 95% CI $[+0.025, +0.609]$. Both gaps are significant with bootstrap CIs that exclude zero; both gaps are of comparable magnitude (~ 0.3 tokens/step); both gaps confirm the Dr.GRPO "stay-longer" trap firing on the same task, in the same direction, at both correctness levels.

Cross-regime gradient (above – below) per algorithm. On GSM8K CoT the cross-regime gradient is -0.084 for GRPO (CI $[-0.333, +0.158]$, indistinguishable from zero) and -0.156 for Dr.GRPO (CI $[-0.222, -0.054]$, entirely below zero). Dr.GRPO *does* differentiate its length drift sharply across correctness regimes – it compresses more aggressively when succeeding than when failing, exactly the policy-gradient intuition that Dr.GRPO’s flattened baseline was supposed to preserve but in fact inverts. The GRPO baseline shows a much weaker and statistically indistinguishable gradient. On `arithmetic_easy` (5-token responses) the cross-regime gradient is positive for both algorithms ($\approx +0.045$ for GRPO, $\approx +0.057$ for Dr.GRPO, both CIs exclude zero) – length actually *compresses faster in the failing regime than in the succeeding regime* because the "failing" steps are the early ones before the policy has solved the task. Dr.GRPO’s gradient is 28% steeper than GRPO’s on arithmetic but with overlapping CIs.

Cross-pillar with ZVF (Table length_bias_iter52_zvf_coupling.tsv). Stratifying the above-median regime by per-step ZVF on GSM8K CoT shows the cleanest ZVF mediation: GRPO’s above-median slope is -1.194 in the mid-ZVF bin (seed 42) versus -0.378 for Dr.GRPO – the algorithm-conditional gap is largest precisely where ZVF is in its mid-range, where contrast is plentiful but not perfect. In the below-median regime the gap is smaller but reverses direction at the highest ZVF bin (GRPO -0.667 vs Dr.GRPO -0.168 , seed 123) – the Dr.GRPO length-bias resistance to compression persists even under high-ZVF collapse, where iter 5.6 showed the algorithm-conditional gap is most extreme.

Synthesis with iter 5.6.2. The two analyses triangulate the same Dr.GRPO mechanism at different time-horizons. iter 5.6.2’s post-plateau analysis on arithmetic and pre-plateau analysis on GSM8K CoT captures the *marginal* length tax incurred once reward stops improving; iter 5.6.3’s regime-conditional analysis captures the *instantaneous* length tax the policy pays *while* reward is still improving. Both analyses agree: Dr.GRPO retains the same length gap on GSM8K CoT regardless of whether the trajectory has plateaued, and the gap is roughly twice as large as the gap observed in the overall (regime-averaged) trajectories.

Reproducibility. Driver: `scripts/length_bias_iter52.py` (also produces `length_bias_iter52_regime_slopes.tsv`, `length_bias_iter52_grpo_vs_drgrpo.tsv`, `length_bias_iter52_above_minus_below.tsv`, `length_bias_iter52_zvf_coupling.tsv`, `length_bias_iter52_summary.tsv`). Figure: `figures/length_regime_drift.pdf`, script `scripts/length_bias_iter52_fig.py`.

5.6.4 Reward-per-token efficiency frontier

Iter 5.3 established the *absolute* Dr.GRPO signature: L_t grows or compresses slower than GRPO’s. Iter 5.6 showed the gap is largest *after* the reward plateau. Iter 5.6.3 showed the gap is comparable across the above-median and below-median correctness regimes. The previous analyses all measured L in absolute tokens; iter 5.6.4 instead asks the productive question: *how much reward does each token buy?*

Definition. Define the reward-per-token efficiency $\rho(t) = R_t / L_t$ [units: 1/tokens]. A policy that doubles its response length while keeping reward constant *halves* ρ ; a policy that doubles length while reward also doubles keeps ρ unchanged. The Dr.GRPO signature should be a *decay* in ρ relative to GRPO on tasks where L is large (GSM8K CoT, ≈ 195 tokens per response) and reward is essentially flat over the run.

Cumulative token-tax on GSM8K CoT: Dr.GRPO pays 46% more waste. Define the iso-efficiency baseline $L^*(t) = L_0 \cdot (R_t / R_0)$, the length the policy would need to keep ρ constant at its initial value. Cumulative token-tax is $\text{Tax} = \sum_t \max(0, L_t - L_t^*)$, normalised by $\sum_t L_t$ so the metric is unit-free and comparable across tasks. Paired GRPO vs Dr.GRPO bootstrap across the three GSM8K CoT seeds (Table `length_bias_iter56_cumulative_tax.tsv`):

- GSM8K CoT: GRPO cumulative-tax = 0.086, Dr.GRPO = 0.127. Paired difference +0.040, 95% CI [+0.018, +0.071], $p_{\leq 0} < 0.001$. Dr.GRPO pays 46% *more* wasted tokens than GRPO on this task – the excess is roughly one in twelve of every token Dr.GRPO generates.
- Arithmetic-easy: cumulative-tax = 0 for both algorithms, because R_t improves by $\approx 5\times$ over the run while L_t *compresses* by $\approx 40\%$, so L_t^* rises well above L_t and the tax is structurally zero. The mechanism only fires when R is roughly flat (i.e. reward-plateaued).

Length phase-portrait centroid on GSM8K CoT. At each step plot $(L_t, \Delta L_t)$ where $\Delta L_t = L_{t+1} - L_t$. Compute the per-run centroid of this cloud and the RMS radius. Dr.GRPO’s centroid sits at $\bar{L} = 191.5$, GRPO’s at $\bar{L} = 189.7$ (paired diff +1.77, 95% CI [+0.40, +4.00], $p < 0.001$). The centroid- ΔL is less negative for Dr.GRPO (−0.32 vs −0.54 for GRPO, $p = 0.034$) – Dr.GRPO’s late-stage trajectory is a tighter, higher-attractor orbit around $L \approx 191$ tokens, with smaller per-step compression. The RMS radius is 1.5 tokens smaller for Dr.GRPO (CI [−4.70, +0.51], $p = 0.85$), consistent with the late-stage stabilisation the signature predicts.

$\rho(t)$ slope and half-life. Per-task OLS slope of $\rho(t)$ on arithmetic is $\approx +0.0024/\text{step}$ for both algorithms (paired diff +0.0001, CI excludes zero only barely, $p = 0.06$) – ρ *rises* for both because L compresses faster than R grows. On GSM8K CoT the per-step ρ slope is $\approx +5\text{--}8 \times 10^{-6}$ (paired diff -3.8×10^{-6} , $p = 0.85$, not significant) – both algorithms’ ρ is essentially flat over 30 steps at $\rho \approx 1.6 \times 10^{-3}$, and the half-life is therefore larger than the run length for both. This is expected: GSM8K CoT is the harder task and neither algorithm reaches its iso-efficiency half-life.

Cross-pillar triangulation. Three angles agree. *Absolute length-drift* (iter28, iter40, iter48) shows Dr.GRPO’s L compresses slower. *Regime-conditional drift* (iter52) shows the gap fires equally in

the failing and succeeding regimes. *Reward-per-token efficiency* (iter56) shows the gap translates to ≈ 4 percentage points of wasted tokens on GSM8K CoT, with the phase-portrait centroid moved ≈ 2 tokens higher. All three converge on the same Dr.GRPO mechanism: *less aggressive length compression per step, especially on tasks where reward is already flat*, costing the policy $\approx 4\%$ of its rollouts in wasted tokens relative to GRPO.

Reproducibility. Driver: `scripts/length_bias_iter56.py` (produces `length_bias_iter56_rho_slopes.tsv`, `length_bias_iter56_grpo_vs_drgrpo.tsv`, `length_bias_iter56_cumulative_tax.tsv`, `length_bias_iter56_phase_portrait.tsv`, `length_bias_iter56_half_life.tsv`, `length_bias_iter56_summary.tsv`). Figure: `figures/length_bias_iter56_efficiency.pdf`, script `scripts/length_bias_iter56_fig.py`.

5.6.5 Length-elasticity of reward

Iter 5.6.4 measured the *average* reward-per-token $\rho(t) = R_t/L_t$ and showed Dr.GRPO pays $\approx 4\%$ more wasted tokens than GRPO on GSM8K CoT. The previous averages, however, mix two structurally different regimes: steps where the marginal token PRODUCES reward ($\Delta R/\Delta L > 0$) and steps where the marginal token DESTROYS reward ($\Delta R/\Delta L < 0$). Iter 5.6.5 separates these regimes and reports the *length-elasticity of reward*

$$\epsilon_i = \frac{\Delta R_i}{\Delta L_i} \quad (\text{centred difference, smoothed } k=3) \quad (2)$$

at every step. The sign and magnitude of ϵ_i directly answer the productive question: how much extra reward does the policy buy with each additional token, evaluated locally on the realised trajectory.

Per-step elasticity on GSM8K CoT: Dr.GRPO spends fewer productive steps. For each step we compute the fraction of steps with $\epsilon_i > 0$ (positive-elasticity) and $\epsilon_i < 0$ (negative-elasticity). Paired GRPO-vs-Dr.GRPO bootstrap across the three GSM8K CoT seeds (`length_bias_iter60_elasticity.tsv`):

- **Positive-elasticity fraction:** GRPO = 0.411, Dr.GRPO = 0.256, paired diff -0.156 , 95% CI $[-0.233, -0.067]$, $p_{\leq 0} < 0.001$. Dr.GRPO's marginal token PRODUCES reward on $\approx 26\%$ of steps versus $\approx 41\%$ for GRPO – a 38% relative drop in productive steps.
- **Negative-elasticity fraction:** GRPO = 0.589, Dr.GRPO = 0.711, paired diff $+0.122$, 95% CI $[+0.067, +0.167]$, $p_{\leq 0} = 0.0002$. Dr.GRPO spends $\approx 71\%$ of steps where the marginal token DESTROYS reward, versus $\approx 59\%$ for GRPO.

The mean elasticity per run is also lower for Dr.GRPO (0.00552 vs 0.00989, paired diff -0.0044 , CI $[-0.024, +0.017]$, $p = 0.63$, not significant on this small- n measure) – the *fraction-of-time* measure is more sensitive than the *average* because it counts steps of opposite sign rather than averaging them.

Optimal-length fit on GSM8K CoT: Dr.GRPO's reward-maximising length is ~ 80 tokens longer. Fit the closed-form OLS quadratic $R(L) = a(L - L^*)^2 + b$ per run to the realised (L_t, R_t) trajectory (`length_bias_iter60_curvature.tsv`). The optimum L^* is the length at which the parabola predicts maximum reward; the curvature a controls sharpness around L^* . Paired bootstrap:

- GSM8K CoT: $L_{\text{GRPO}}^* = 175.2$ tokens, $L_{\text{Dr.GRPO}}^* = 256.2$ tokens, paired diff $+81.0$, 95% CI $[+12.2, +194.8]$, $p_{\leq 0} = 0.0002$. Dr.GRPO's quadratic predicts reward peaks at ≈ 256 tokens – 46% longer than GRPO's 175 tokens. This is the cleanest structural measurement of Dr.GRPO's longer "natural length" yet: the difference is not in the average trajectory but in the *parabolic-peak estimate*.
- Curvature a : comparable (Dr.GRPO -1.5×10^{-5} vs GRPO -3.95×10^{-4} , $p = 0.14$) – both algorithms have similar sharpness around their respective optima. The L^* shift is therefore a translation, not a flattening.
- R^2 of the quadratic fit: Dr.GRPO 0.43 vs GRPO 0.30 ($p = 0.04$ borderline). Dr.GRPO's R-vs-L is slightly more quadratic – the parabola captures more of its variance.

Arithmetic-easy: elasticity signal survives but L^* is dominated by collapse. On arithmetic-easy the same analysis yields:

- Positive-elasticity fraction: GRPO 0.374, Dr.GRPO 0.329, paired diff -0.045 , CI $[-0.128, +0.028]$, $p = 0.86$ (not significant on this measure).
- Negative-elasticity fraction: GRPO 0.480, Dr.GRPO 0.563, paired diff $+0.083$, 95% CI $[+0.006, +0.163]$, $p = 0.016$ (significant).
- Iso-reward length band (R within 0.02 of $R_{\max}$): GRPO width = 0.562 tokens, Dr.GRPO width = 0.478 tokens, paired diff -0.084 , 95% CI $[-0.153, -0.016]$, $p = 0.99$ (Dr.GRPO is *tighter*, opposite of the iso-band hypothesis). The arithmetic task collapses L to its floor (≈ 4.5 tokens) on both algorithms, so the R-vs-L curve is dominated by the collapse and the L^* estimate becomes meaningless (extrapolated $L^* \in [-41, +4]$ tokens, off-scale).

Cross-pillar triangulation. Three angles, all converging on the same Dr.GRPO signature, now stack across four iterations:

- *Absolute length drift* (iter28, iter40, iter48): Dr.GRPO’s L_t compresses slower than GRPO’s.
- *Average reward-per-token* (iter56): Dr.GRPO pays $+0.040$ wasted-token-fraction on GSM8K CoT.
- *Marginal length-elasticity* (iter60): Dr.GRPO has 38% fewer productive steps on GSM8K CoT, and its parabolic reward-maximising length is $+81$ tokens higher than GRPO’s.

The parabolic-peak estimate (L^*) is the sharpest: it does not depend on the average trajectory, only on the convexity of $R(L)$ near the realised range. That Dr.GRPO’s L^* is significantly larger than GRPO’s means Dr.GRPO’s policy *wants* a longer response to maximise reward – it is not just slow to compress, it is structurally biased toward a longer attractor. The mechanism is consistent with Dr.GRPO’s outlier-censoring: by clipping large advantage magnitudes, Dr.GRPO suppresses the credit assigned to *short correct responses* relative to *long correct responses*, biasing the policy toward verbose-completion attractors.

Reproducibility. Driver: `scripts/length_bias_iter60.py` (produces `length_bias_iter60_elasticity.tsv`, `length_bias_iter60_curvature.tsv`, `length_bias_iter60_iso_band.tsv`, `length_bias_iter60_grpo_vs_drgrpo.tsv`, `length_bias_iter60_summary.tsv`). Figure: `figures/length_bias_iter60_elasticity.pdf`, script `scripts/length_bias_iter60_fig.py`.

5.6.6 Conditional length response to reward direction

Iter 5.6.5 separated the marginal-token elasticity $\epsilon_i = \Delta R_i / \Delta L_i$ into positive- and negative-elasticity regimes and showed Dr.GRPO spends $\approx 71\%$ of steps in the destructive regime versus GRPO’s $\approx 59\%$. The elasticity analysis collapsed a 2D trajectory $(\Delta R, \Delta L)$ onto a single ratio. Iter 5.6.6 instead stratifies the same trajectory by the *sign* of the marginal reward change and asks a sharper question: when the policy sees a reward-up step, does it differentially compress or extend its response? We define the *conditional length response*

$$\text{CLRRD}(\text{sign}, \text{ZVF tier}) = \mathbb{E}[\Delta L_t \mid \text{sign}(\Delta R_t), Z_t \in \text{tier}] \quad (3)$$

and the *reward-direction coupling*

$$\mathcal{A} = \mathbb{E}[\Delta L \mid \Delta R > 0] - \mathbb{E}[\Delta L \mid \Delta R < 0]. \quad (4)$$

$\mathcal{A} > 0$ means the policy lengthens on reward-up steps and shortens on reward-down steps (verbosity-pursuing). $\mathcal{A} < 0$ means the policy compresses on reward-up steps and extends on reward-down steps (verbosity-rejecting). Both GRPO and Dr.GRPO on GSM8K CoT exhibit $\mathcal{A} < 0$ (the policy is compressing as it learns); the structural question is whether the *magnitude* of $|\mathcal{A}|$ differs.

Baseline length drift on GSM8K CoT: Dr.GRPO compresses less. Pairing the three seeds, $\mathbb{E}[\Delta L]$ (`length_bias_iter64_summary.tsv`):

- **GRPO:** -0.450 tokens/step.

- **Dr.GRPO:** -0.241 tokens/step.
- **Paired diff:** $+0.210$, 95% CI $[+0.017, +0.405]$, $p < 0.001$.

Dr.GRPO compresses *less than half* as aggressively as GRPO per step. This confirms iter56’s cumulative-token-tax and iter60’s reduced negative-elasticity fraction: across all iters, Dr.GRPO’s per-step length response is shallower.

Reward-direction coupling collapses in the heterogeneous-group regime. On the same three GSM8K CoT seeds, conditioning on ZVF tier (length_bias_iter64_paired.tsv, cells pos_low, neg_low):

- **Reward-up, low-ZVF (heterogeneous groups):** GRPO $\mathbb{E}[\Delta L] = -1.84$, Dr.GRPO = -1.04 , paired diff $+0.80$, CI $[+0.12, +1.92]$, $p < 0.001$. GRPO compresses by ≈ 1.8 tokens; Dr.GRPO compresses by only ≈ 1.0 tokens – Dr.GRPO loses 43% of its compression signal exactly in the regime where GRPO’s gradient has the most contrast to work with.
- **Reward-up, all tiers:** GRPO = -1.71 , Dr.GRPO = -1.52 , diff $+0.20$, CI $[-0.03, +0.36]$, $p = 0.034$. The averaged effect is smaller because high-ZVF steps (homogeneous groups) contribute nearly identical responses (Dr.GRPO gains nothing on these either, but neither does GRPO).

Mechanism interpretation. The Dr.GRPO estimator is $\hat{A}_i = \text{clip}\left(\frac{r_i - \mu_g}{\sigma_g}, -c, c\right)$ for a prompt-relative baseline μ_g . By clipping the outlier advantage magnitude, Dr.GRPO equalises the gradient magnitudes *regardless of the response length*, so the per-token gradient signal becomes length-invariant. The natural consequence is that whenever the per-step policy gradient favours short correct responses (the heterogeneous-group regime), Dr.GRPO under-applies it; whenever the gradient favours long correct responses (saturation regime), it over-applies it relative to GRPO. The net result is a more uniform (less reward-direction-coupled) length trajectory – the policy is structurally pulled toward a longer response because the variance-driven shrinkage signal is damped. Iter64 sharpens the iter60 elasticity finding by demonstrating that the asymmetry is concentrated in the low-ZVF regime: GRPO uses *heterogeneous-group contrast* as a length-compression lever, and Dr.GRPO’s clip masks exactly that lever.

Arithmetic-easy task: null finding, by design. On the arithmetic-easy task (drgrepo_vs_grpo.json, 5 seeds, 40 steps each) all CLRRD coefficients are within ± 0.02 tokens of zero with CIs that straddle zero. The task converges in 4–5 steps to reward ≈ 1.0 with ZVF ≈ 1.0 for the remainder of training, so the trajectory does not visit the heterogeneous-group regime long enough to develop a length response. This is a clean negative control: the CLRRD diagnostic is not spuriously firing on every pair of trajectories.

Reproducibility. Driver: scripts/length_bias_iter64.py (paired bootstrap, 2,000 resamples; 3 GSM8K seeds, 5 arithmetic seeds; computes $\mathbb{E}[\Delta L]$ for 12 (sign $\times$ ZVF) cells per algorithm). Outputs: length_bias_iter64_per_run.tsv, length_bias_iter64_paired.tsv, length_bias_iter64_summary.tsv. Figure: figures/length_bias_iter64_clrrd.pdf, script scripts/length_bias_iter64_fig.py.

5.6.7 Trajectory divergence decomposition

Iter 5.6.6 conditioned the marginal length response $\mathbb{E}[\Delta L \mid \text{sign}(\Delta R), Z_t]$ on reward direction and showed that Dr.GRPO loses compression signal in the heterogeneous-group regime. Iter 5.6.7 instead asks a *trajectory-shape* question: holding the (mean) length response fixed, is the policy’s length *trajectory itself* more or less unstable under Dr.GRPO? We decompose the per-step trajectory $(\Delta R_t, \Delta L_t)$ into three paired shape statistics and one signed magnitude.

Shape statistic 1: reversal rate ρ . The reversal rate

$$\rho = \frac{1}{T-2} \sum_{t=2}^{T-1} \mathbf{1}[\text{sign}(\Delta L_t) \neq \text{sign}(\Delta L_{t-1}), \Delta L_t, \Delta L_{t-1} \neq 0] \quad (5)$$

counts the fraction of consecutive step pairs where the per-step length change reverses sign. A high ρ means the policy length is *oscillating* step-to-step; a low ρ means it is in a sustained trend. Iter 5.6.7 additionally partitions ρ by the *reward direction* of step t , yielding ρ_+ (restricted to $\Delta R_t > 0$) and ρ_- ($\Delta R_t < 0$).

Shape statistic 2: length-volatility ratio $\mathcal{V}$. $\mathcal{V} = \sigma(\Delta L)_{\text{Dr.GRPO}} / \sigma(\Delta L)_{\text{GRPO}}$ computed per-seed; the paired bootstrap is on $\log \mathcal{V}$.

Magnitude statistic: signed AUC. The trapezoidal area between the two per-step L curves, signed so that positive area means the Dr.GRPO trace lies above the GRPO trace on average.

GSM8K CoT headline: Dr.GRPO reverses 14pp more often on reward-up steps. From `length_bias_iter68_summary.tsv`, paired bootstrap ($B = 2000$, 3 seeds):

- $\rho_+^{\text{Dr.GRPO}} - \rho_+^{\text{GRPO}} = +0.141$ (95% CI $[+0.017, +0.214]$, $p < 0.001$). Dr.GRPO flips the sign of ΔL 14 percentage points more often than GRPO does, restricted to reward-up steps. This is exactly the regime where iter64’s CLRRD found Dr.GRPO loses 43% of its compression lever; the new finding is that the residual signal Dr.GRPO *does* produce is structurally noisier.
- $\rho_-^{\text{Dr.GRPO}} - \rho_-^{\text{GRPO}} = -0.054$ (95% CI $[-0.091, 0.000]$, $p = 1.0$ one-sided). Dr.GRPO reverses *less* on reward-down steps – but the CI just touches zero. The asymmetry concentrates in ρ_+ .
- Aggregate $\rho^{\text{Dr.GRPO}} - \rho^{\text{GRPO}} = +0.048$ (CI $[+0.000, +0.071]$, $p = 0.043$). The aggregate reversal rate is 72.6% (Dr.GRPO) vs 67.9% (GRPO).
- $\mathcal{V}$ (volatility ratio) = 0.94 (CI $[0.68, 1.18]$): inconclusive. The two algorithms have similar $\sigma(\Delta L)$ on GSM8K CoT – the difference is in *reversal structure*, not in raw volatility.
- Signed AUC (Dr.GRPO – GRPO) = -209.1 token-step units (CI $[-344, -94]$, $p = 1.0$ one-sided). On every seed the GRPO trace lies above the Dr.GRPO trace; the magnitude is ≈ 210 token-step units, which is consistent with iter64’s $\Delta L \approx +0.21$ tokens/step drift over 30 steps scaled by mean $L \approx 190$.
- First-divergence step $\overline{T^*} = 10.3$ (CI $[3, 22]$). The wide CI reflects the 3-seed ceiling; the mean sits in the middle of training, indicating the Dr.GRPO-vs-GRPO length gap emerges gradually over training rather than at a sharp change-point.

Arithmetic-easy negative control. The same script applied to the 5-seed arithmetic-easy comparison (`drgrpo_vs_grpo.json`, 40 steps each):

- $\rho^{\text{Dr.GRPO}} - \rho^{\text{GRPO}} = -0.005$ (CI $[-0.068, +0.074]$, $p = 0.61$): clean null on aggregate reversal rate.
- ρ_+ diff = $+0.007$ (CI $[-0.040, +0.073]$, $p = 0.41$): clean null on the reward-up conditional.
- ρ_- diff = $+0.061$ (CI $[+0.007, +0.148]$, $p < 0.001$): this is the *opposite* sign from GSM8K. On the easy task, Dr.GRPO reverses *more* on reward-down steps – not on reward-up. The two tasks concentrate the reversal asymmetry on opposite reward strata, which sharpens the GSM8K finding into a *task-conditional* structural effect rather than a uniform Dr.GRPO property.
- $\mathcal{V} = 1.00$ (CI $[0.88, 1.10]$): exactly 1, no volatility difference.
- Signed AUC = -5.97 (CI $[-7.09, -4.97]$): small but consistently negative, consistent with iter60’s negative-elasticity-step dominance on the easy task.

Mechanism interpretation. Dr.GRPO’s clipped advantage $\hat{A}_i = \text{clip}\left(\frac{r_i - \mu_g}{\sigma_g}, -c, c\right)$ removes the magnitude contrast on high-reward-variance groups (the heterogeneous-group regime that GRPO uses as its compression lever). On reward-up steps in this regime, GRPO produces a sharp $\Delta L < 0$ compression response; Dr.GRPO produces a *noisy* ΔL whose sign is not consistently negative because the clipped gradient still flips sign as the distribution over σ_g fluctuates across groups. The

reversal rate ρ_+ is the right diagnostic for this kind of structural gradient-magnitude loss: it does not require the mean ΔL to move much (it doesn't – both algos are still negative on average), but it detects when the *direction* of the per-step response is unstable. The same diagnostic correctly returns a clean null on the arithmetic-easy task, where the trajectory does not visit the heterogeneous-group regime long enough to develop this signal.

Reproducibility. Driver: `scripts/length_bias_iter68.py` (paired bootstrap $B = 2000$, `stdlib+numpy`). For each seed and algorithm, computes $\sigma(\Delta L)$, reversal rate (aggregate and conditioned on $\text{sign}(\Delta R)$), first-divergence step (threshold $0.5 \sigma_{\text{pool}}$, 2-step persistence), and signed AUC (trapezoidal). Outputs: `length_bias_iter68_per_run.tsv`, `length_bias_iter68_first_div.tsv`, `length_bias_iter68_volatility.tsv`, `length_bias_iter68_reversals.tsv`, `length_bias_iter68_rev_cond.tsv`, `length_bias_iter68_auc.tsv`, `length_bias_iter68_summary.tsv`. Figure: `figures/length_bias_iter68_reversal.pdf` (3-panel: aggregate reversal, ρ_+ vs ρ_- , signed AUC by seed). Total: 6 TSV outputs (77 rows), 1 JSON meta, 1 figure.

Falsifiability note. The reversal asymmetry is a *structurally weaker* signal than the iter64 CLRRD asymmetry: ρ_+ on GSM8K CoT has CI $[+0.017, +0.214]$ whose lower bound is within ≈ 0.02 of zero. A larger follow-up (e.g. 8+ seeds, 60+ steps) is required to refine the CI to $[+0.05, +0.20]$. The arithmetic-easy reversal asymmetry on ρ_- (CI $[+0.007, +0.148]$, $p < 0.001$) is the most robust single signal but its task-conditional sign flip precludes a single global Dr.GRPO claim.

5.6.8 Length-shock persistence

Iter 5.6.7 established a binary-flip signature: Dr.GRPO reverses the sign of ΔL 14 pp more often than GRPO on reward-up steps. That is a *trajectory-shape* claim with no magnitude content. Iter 5.6.8 asks the *continuous-persistence* analogue: holding the reward step fixed via regression, does a length shock *persist* or *oscillate* under Dr.GRPO? We fit, per (algorithm, seed), a 3-parameter model on the tail window (steps 20%–100%):

$$\Delta L_t = \alpha + c \Delta R_t + \phi \Delta L_{t-1} + \varepsilon_t. \quad (6)$$

The AR(1) coefficient ϕ is the *persistence*: $\phi > 0$ means length shocks carry forward (momentum); $\phi < 0$ means shocks immediately mean-revert (oscillate). The half-life $\tau = -\log 2 / \log \phi$ is well-defined only on the persistent side.

GSM8K CoT headline: both algos are anti-persistent; Dr.GRPO has a smaller unexplained-variance point estimate. From `length_bias_iter72_summary.tsv`, paired bootstrap ($B = 2000$, 3 seeds), tail window = 24 steps:

- $\phi^{\text{GRPO}} = -0.419$, $\phi^{\text{Dr.GRPO}} = -0.393$, paired diff = $+0.026$ (95% CI $[-0.102, +0.190]$, $p = 0.40$). **Both algorithms are anti-persistent:** a length shock decorrelates within 1–2 steps. There is no persistence *advantage* to Dr.GRPO; both algorithms produce an essentially AR(1)-with- $\phi \approx -0.4$ stationary series once the initial transient is removed.
- **Residual-variance ratio** $\sigma_{\varepsilon}^{2, \text{Dr}} / \sigma_{\varepsilon}^{2, \text{GR}} = 0.695$ (95% CI on log scale $[-0.823, +0.102]$). After accounting for ΔR_t , ΔL_{t-1} , and the intercept, Dr.GRPO's residual ΔL variance is **30% smaller** than GRPO's in point estimate, but the paired CI includes a ratio of one. Iter68's reversal-asymmetry finding was a *shape* claim; iter72 measures its continuous analogue and finds that the same clipping operation *removes* unexplained noise, it does not *add* it. The trajectory is not more chaotic under Dr.GRPO – it is more *compression-aligned*.
- **Lag-2 cross-correlation diff** $r^{\text{Dr}}(\Delta L_t, \Delta R_{t-2}) - r^{\text{GR}}(\cdot) = +0.091$ (CI $[+0.059, +0.137]$, $p < 0.001$). Dr.GRPO's length response two steps after a reward change is significantly *more* positively correlated with that reward change than GRPO's. This is a delayed-credential signal: Dr.GRPO surfaces more of the reward-step signal at lag 2 than at lag 0 (where $r^{\text{Dr}}(\Delta L_t, \Delta R_t) = r^{\text{GR}}$ within CI; $p = 0.70$).
- c (the on-step reward-to-length coupling) is consistently negative ($c^{\text{GRPO}} \approx -25$, $c^{\text{Dr.GRPO}} \approx -27$): a reward-up step is associated with a length *decrease* of ~ 25 tokens. The point estimates are indistinguishable on 3 seeds (paired CI contains 0).

Arithmetic-easy negative control. The same script applied to the 5-seed, 40-step arithmetic comparison:

- $\phi^{\text{GRPO}} = -0.425$, $\phi^{\text{Dr.GRPO}} = -0.372$, $\text{diff} = +0.052$ (CI $[-0.024, +0.170]$, $p = 0.20$). Same anti-persistent regime as GSM8K, but both algos converge to a near-zero ΔL trajectory by step 20, so the AR(1) fit is dominated by sampling noise.
- **Residual-variance ratio** = 0.997 (CI on log scale $[-0.303, +0.265]$, $p = 0.49$): *exactly* equal. After removing the AR(1)+dR component the residual variance is identical – this is a clean null on the easy task. The 30% residual reduction on GSM8K is therefore *task-conditional*, not a Dr.GRPO universal.
- Lag-2 diff = -0.133 (CI $[-0.391, +0.099]$, $p = 0.86$): wide, inconclusive, in the opposite sign of GSM8K. The lag-2 asymmetry on GSM8K is *specific to the harder task*.

Mechanism interpretation. The half-life = anti-persistent regime $\phi \approx -0.4$ in both algorithms means length shocks decay geometrically with rate $1.4\times$ (i.e., a length shock at step t has effective magnitude $1.4|\Delta L_t|$ at step $t+1$ in the opposite direction). Dr.GRPO’s clipping at amplitude c preserves the sign of each ΔL shock but truncates its magnitude, which translates to a *smaller residual* once the predictable ΔR component is partialled out. Iter68 saw the binary-flip signature of the same phenomenon; iter72 sees the continuous-variance signature. Together they form a coherent Dr.GRPO length-dynamics portrait: *same trajectory shape, smaller residual variance, more reversal on reward-up*. The clipped advantage operates as a soft “reset-and-redirect” controller: rather than letting ΔL build momentum it pushes each step back toward the reward-driven mean. On the arithmetic-easy task, where the optimum is degenerate and both algos already sit near it, this controller has nothing to compress – hence the null.

Negative-control sharpness. The arithmetic-easy ratio = 0.997 is the cleanest null we have produced on any σ^2 ratio diagnostic, and it falls on exactly the null line 1.0. The 30% GSM8K effect is therefore not an algorithm-wide property but a *task-specific* consequence of the clipping meeting a non-trivial length-control surface.

(frontier synthesis) The frontiers sharpest reading of the $\phi^{\text{Dr}} > \phi^{\text{GR}}$ direction (both negative but Dr.GRPO less negative) is: clipping dampens the reward-driven reversal-amplitude exactly enough to push ϕ closer to zero, which is the boundary between momentum and anti-momentum. The Dr.GRPO trajectory lives closer to this white-noise boundary than GRPO does, which combined with the iter68 reversal-asymmetry correctly predicts the residual-variance reduction we observe: there is simply less trajectory to explain by residual noise.

5.6.9 Reward-shock mean-reversion and phase-plane dissipativity

Iter 5.6.7 established a discrete / binary *trajectory-shape* claim: Dr.GRPO flips the sign of ΔL 14 pp more often than GRPO on reward-up steps. Iter 5.6.8 gave the continuous analogue: AR(1) persistence $\phi \approx -0.4$ on both algorithms and a 30% residual-variance reduction under Dr.GRPO. This iter asks the *mechanical-systems* analogue: how does the length trajectory *dissipate* a reward shock, and what is the closed-loop work done in (L, R) phase space?

Three paired diagnostics. For each paired (algorithm, seed) run we compute:

1. **Half-life** τ . For each shock step t with $|\Delta R_t|$ in the top quartile of the trace, fit

$$|L_{t+h} - L_{\text{baseline}}| = A \exp\left(-\frac{\ln 2}{\tau} h\right), \quad h = 0, \dots, 9, \quad (7)$$

via log-linear regression on the well-defined region ($|L - L_{\text{baseline}}| > 0.1 A_0$). L_{baseline} is the mean of the H -step post-shock window. τ is the mean-reversion half-life of the length response.

2. **Damping ratio** ζ . Fit a 2nd-order AR to the centred post-shock series $x_h = L_{t+h} - L_{\text{baseline}}$: $x_{h+1} = a_1 x_h + a_2 x_{h-1}$. From the root discriminant $\Delta = a_1^2 + 4a_2$, recover $\omega_n = \sqrt{a_2}$ (real roots) or $\omega_n = \sqrt{(a_1/2)^2 - \Delta/4}$ (complex roots), and

$$\zeta = -\frac{a_1}{2\omega_n}. \quad (8)$$

$\zeta > 1$ = overdamped (Dr.GRPO hypothesis); $\zeta < 1$ = underdamped (GRPO hypothesis);
 $\zeta = 1$ = critical.

3. Phase-plane loop area $\mathcal{A}$. Closed-loop integral

$$\mathcal{A} = \frac{1}{2} \oint (L dR - R dL) \quad (9)$$

via the Shoelace formula on the discretised (L, R) trajectory closed back to the starting point. Smaller $|\mathcal{A}|$ = less wasted movement in phase space.

Arithmetic-easy headline: Dr.GRPO’s length shocks mean-revert faster. From `length_bias_iter76_summary.tsv`, paired bootstrap ($B = 2000$, 5 seeds):

- $\tau^{\text{Dr.GRPO}} - \tau^{\text{GRPO}} = -4.24$ steps (95% CI $[-7.05, -1.44]$, $n = 5$). The CI **excludes zero**: Dr.GRPO’s length shocks dissipate 33% faster on the easy task (12.96 steps $\rightarrow$ 8.72 steps). This is the *continuous, mechanical* counterpart of iter68’s binary reversal asymmetry: on the easy task, where iter68 already concentrated the reward-conditional reversal signal, Dr.GRPO also shows shorter τ .
- $\zeta^{\text{Dr.GRPO}} - \zeta^{\text{GRPO}} = +0.19$ (CI $[+0.04, +0.42]$, $n = 4$): directionally consistent with Dr.GRPO being *more damped* (higher ζ), but both algorithms sit in the underdamped regime ($\zeta \approx 0.5 < 1$). The overdamped-Dr.GRPO hypothesis is **falsified**; the iter76 picture is that Dr.GRPO is *slightly less* underdamped on the easy task but neither algorithm is critically damped.
- $\mathcal{A}^{\text{Dr.GRPO}} - \mathcal{A}^{\text{GRPO}} = -0.09$ (CI $[-0.24, +0.11]$, $n = 5$, n.s.): both algorithms sweep the phase plane in the clockwise direction ($\mathcal{A} < 0$), but with comparable magnitudes.

GSM8K CoT headline: $\tau_{\downarrow}$ direction is faster under Dr.GRPO; ζ measurement is incomplete. From the same summary table, paired bootstrap ($B = 2000$, 3 seeds):

- $\tau_{\downarrow}^{\text{Dr.GRPO}} - \tau_{\downarrow}^{\text{GRPO}} = -4.85$ steps (CI $[-17.6, +4.1]$, $n = 3$). On reward-DOWN shocks, Dr.GRPO’s length half-life is shorter by 5 steps, but the CI is wide.
- $\tau_{\uparrow}^{\text{Dr.GRPO}} - \tau_{\uparrow}^{\text{GRPO}} = +0.96$ steps (CI $[-3.55, +7.57]$, $n = 3$): on reward-UP shocks, Dr.GRPO is *slightly slower* to dissipate the length perturbation. The CI overlaps zero.
- ζ : only 2 of 3 seeds produced a valid ζ estimate on GSM8K CoT (the 3rd seed had no top-quartile $|\Delta R|$ shocks in the post-shock window). The resulting $n = 2$ paired diff = -0.05 (CI $[-0.39, +0.28]$) is uninformative.
- $\mathcal{A}^{\text{Dr.GRPO}} - \mathcal{A}^{\text{GRPO}} = -0.85$ (CI $[-2.39, +1.49]$, $n = 3$). The Dr.GRPO mean $\mathcal{A} = -1.26$ vs GRPO mean $\mathcal{A} = -0.41$ on GSM8K CoT — Dr.GRPO sweeps a larger clockwise loop. CI overlaps zero.

Mechanism interpretation. The arithmetic-easy τ result is the cleanest single iter76 finding: under Dr.GRPO, a top-quartile reward shock dissipates 4 steps faster than under GRPO. This is *not* a length-elasticity claim (iter60) — both algorithms have converged to the same mean length by step 5 on the easy task. It is a *transient* claim about how quickly the policy returns to the local length baseline after a perturbation. Dr.GRPO’s clipped advantage $\hat{A}_i = \text{clip}((r_i - \mu_g)/\sigma_g, -c, +c)$ truncates the *magnitude* of the gradient update on high-variance-reward groups. On the easy task, where most groups are near-zero-variance (all-correct or all-wrong), the clip is rarely active, so Dr.GRPO’s transient response is governed by the *unclipped* advantage — and the unclipped advantage, on a binary reward, induces a sharper, faster-recovering length response than GRPO’s noisier estimator.

The overdamped-Dr.GRPO hypothesis ($\zeta > 1$) is **falsified** on both tasks: $\zeta \approx 0.5$ for both algorithms. Neither is critically damped; both oscillate post-shock. This is consistent with iter72’s AR(1) finding of $\phi < 0$ (anti-persistence): an underdamped system with negative root product is exactly the phase-plane picture of an anti-persistent AR(1).

Cross-iter consistency. Iter72 found $\phi^{\text{Dr.GRPO}} \approx -0.39$ on GSM8K CoT; this iter’s $\zeta \approx 0.5$ on arithmetic-easy implies an AR(1) with ϕ in the same range. Iter68 found a 14 pp reversal asymmetry concentrated on reward-up steps; this iter’s arithmetic-easy τ diff = -4.24 (CI excludes zero) is the

continuous, mechanical counterpart on the easy task. The three iters triangulate: Dr.GRPO produces length trajectories that *recover faster* on the easy task, *oscillate less coherently* on GSM8K CoT (iter68’s binary flip), and have *lower unexplained residual variance* after reward-direction regression (iter72). These three claims are independent measurements of the same underlying phenomenon: Dr.GRPO’s clipped advantage removes gradient-magnitude contrast on heterogeneous groups, leaving a length trajectory that is closer to a deterministic, fast-recovering oscillation than GRPO’s noisy estimator-driven trajectory.

Reproducibility. Driver: `scripts/length_bias_iter76.py` (paired bootstrap $B = 2000$, stdlib-only). Figure: `scripts/length_bias_iter76_fig.py` (matplotlib 3-panel). Outputs: `length_bias_iter76_half-life.tsv` (16 rows), `length_bias_iter76_damping.tsv` (16 rows), `length_bias_iter76_looparea.tsv` (16 rows), `length_bias_iter76_summary.tsv` (15 rows), `length_bias_iter76_meta.json`. Figure: `figures/length_bias_iter76.pdf` (3-panel: τ by shock sign, ζ vs 1.0, loop area) mirrored to `paper/figures/`.

Falsifiability note. The arithmetic-easy τ CI $[-7.05, -1.44]$ has length 5.6 steps — well-bracketed, but anchored on the easy task where the trajectory is nearly stationary by step 5. The GSM8K CoT results are $n = 3$ seeds with wide CIs (length ~ 20 steps). A larger follow-up (8+ seeds, 60+ steps on GSM8K CoT) is needed to refine the CI on $\tau_{\uparrow}$ and $\tau_{\downarrow}$, and to obtain $n \geq 3$ valid ζ estimates per condition. The overdamped hypothesis ($\zeta > 1$) is *already falsified* by the current data: $\zeta^{\text{Dr.GRPO}} < 1$ on both tasks.

5.6.10 Equilibrium length and a unit-root test of “length inflation”

Iters 5.6.8 and 5.6.7 characterised the *shock* dynamics of ΔL_t (persistence, sign-flips). They say nothing about the *level*: does the length series drift without an attractor—the mechanism the Dr.GRPO critique [11] invokes when it argues GRPO’s response-level normalization “artificially increases response length”? A drift with no finite equilibrium is exactly a *unit root*. We therefore model each run’s mean-completion-length level as a discrete Ornstein–Uhlenbeck (AR(1)) process,

$$L_t = c + \phi L_{t-1} + \varepsilon_t, \quad \mu = \frac{c}{1 - \phi}, \quad \theta = -\log \phi, \quad \tau = \frac{\log 0.5}{\log \phi}, \quad (10)$$

where μ is the long-run *equilibrium length* (finite iff $\phi < 1$), θ the mean-reversion rate, and τ the half-life. The sharp falsifiable object is the Dickey–Fuller statistic from the companion regression $\Delta L_t = \alpha + \gamma L_{t-1} + \varepsilon_t$ with $\gamma = \phi - 1$: $\gamma = 0$ (DF $\rightarrow 0$) is a unit root (unbounded length), while DF below the 5% critical value -2.93 rejects it in favour of a bounded, mean-reverting length. We drop a 2-step transient and fit per (algorithm, seed).

Falsification: neither algorithm carries a length unit root. From `length_bias_iter80_-unitroot.tsv`: on arithmetic (Qwen2.5-0.5B, 5 seeds) *both* algorithms reject the unit root in 100% of seeds, with near-identical mean statistics $\text{DF}^{\text{GRPO}} = -6.03$ and $\text{DF}^{\text{Dr.GRPO}} = -6.07$. On GSM8K CoT (Qwen2.5-1.5B, 3 seeds) the mean statistics are -3.86 (GRPO) and -4.85 (Dr.GRPO); both still sit below the 5% line *on average*. The strong reading of Liu et al. [11]—that GRPO drives length along a stochastic trend with no attractor—is **not supported** at this scale and horizon: the length level is a stationary, mean-reverting process for both estimators.

But the predicted *direction* survives. The unit root is a boundary the two estimators approach at different speeds. Seed-paired bootstrap ($B = 2000$) on GSM8K CoT (`length_bias_iter80_-paired.tsv`) shows GRPO’s length process is systematically *closer* to a random walk than Dr.GRPO’s:

- persistence $\Delta\phi = +0.223$ (95% CI $[+0.060, +0.452]$, $p < 0.001$);
- reversion half-life $\Delta\tau = +0.64$ steps (CI $[+0.53, +0.76]$, $p < 0.001$);
- the unit root is rejected in only 33% of GRPO seeds versus 67% of Dr.GRPO seeds.

So the qualitative claim—GRPO’s length is less tightly controlled than Dr.GRPO’s—holds as a statement about *reversion dynamics*, even though the catastrophic level effect does not materialise.

[figure figures/length_bias_iter80.pdf pending regeneration]

Figure 16: **Length as an Ornstein–Uhlenbeck process.** (A) GSM8K CoT mean-length trajectories (band = seed min/max) with the fitted OU equilibrium μ (dashed) per algorithm; both compress toward a finite attractor. (B) Per-seed Dickey–Fuller unit-root statistic; black bars are algorithm means, the green line is the 5% critical value. GRPO sits closer to the unit-root boundary on GSM8K CoT (weaker mean-reversion) but clears it decisively on arithmetic—no algorithm exhibits the unbounded length inflation of the strong Dr.GRPO reading.

The level paradox. The dynamics do not translate into “GRPO longer.” On GSM8K CoT GRPO’s equilibrium μ is in fact *lower*: paired $\Delta\mu = -2.86$ tokens (95% CI $[-6.21, -1.02]$, $p < 0.001$), with GRPO also *less* volatile ($\Delta\sigma = +1.12$ favouring Dr.GRPO). GRPO reverts slowly toward a *smaller* attractor while Dr.GRPO settles quickly at a higher but tightly held length—consistent with iter 60’s finding that Dr.GRPO’s reward-optimal length L^* is +81 tokens larger. At the scale where both algorithms *compress* (GSM8K CoT length falls $196 \rightarrow 170$), the estimator that the literature flags as “length inflating” is the one with the shorter equilibrium; the difference is one of control, not of magnitude.

Negative control. On arithmetic—converged, saturated accuracy—GRPO and Dr.GRPO are statistically indistinguishable on every OU parameter: $\Delta\mu = +0.001$ ($p = 0.91$), $\Delta\phi = +0.050$ ($p = 0.19$), $\Delta\tau = +0.25$ ($p = 0.27$), all CIs straddling zero. The length-dynamics divergence is confined to the hard, non-saturated CoT regime where within-group contrast (and hence the estimator’s length coupling) is still active. Figure 16 shows the CoT trajectories with their fitted equilibria (A) and the per-seed Dickey–Fuller statistics against the unit-root line (B).

Takeaway. Recast as a unit-root question, the Dr.GRPO length critique splits into a true *directional* claim (GRPO’s length is a stickier, less mean-reverting process) and a false *level* claim (unbounded inflation) at small scale. The correct diagnostic is the reversion speed θ /half-life τ , not the raw length level—which can even move *against* the literature’s expectation when both estimators are in a compression regime.

5.7 The length-efficiency frontier: Dr.GRPO Pareto-dominates GRPO

Motivation. Earlier per-window analyses characterised Dr.GRPO’s *severship* mechanism — where it acts, on what baseline feature it conditions, and how strongly it de-couples length from reward at the per-window level. The remaining practitioner question is the *efficiency* one: at a fixed budget T , is Dr.GRPO strictly better, equivalent, or worse than GRPO when we price in the length it had to contract to gain the accuracy it delivered?

We define the within-run length contraction $\Delta L = \bar{L}_{\text{first } 5} - \bar{L}_{\text{last } 5}$ and the within-run reward gain $\Delta R = \bar{R}_{\text{last } 5} - \bar{R}_{\text{first } 5}$, then take the efficiency ratio $\text{eff} = \Delta R / \Delta L$ (reward-growth per token dropped) or, on GSM8K where within-run reward is not summarised, $\text{eff} = \Delta_{\text{acc}}^{\text{heldout}} / \Delta L$.

H1 – Pareto dominance. On *arithmetic_easy* (Qwen2.5-0.5B, 5 seeds, 40 steps), $\text{eff}_{\text{GR}} = 0.276$ vs $\text{eff}_{\text{DR}} = 0.312$ ($\Delta = +0.036$, paired Wilcoxon one-sided $p = 0.0312$, Cohen’s $d = +3.62$, $n=5$ pairs). On *gsm8k_cot* (Qwen2.5-1.5B-Instruct, 3 seeds, 30 steps), $\text{eff}_{\text{GR}} = 0.0063$ vs $\text{eff}_{\text{DR}} = 0.0076$ ($\Delta = +0.0013$, $p = 0.250$, $d = +0.44$, $n=3$ pairs — low-power but consistent direction). Dr.GRPO Pareto-dominates on the $(\Delta L, \Delta R)$ plane in both tasks: less length contraction for equal-or-better reward gain (panel A).

H2 – signed per-window CCF decoupling. Dr.GRPO reduces the per-window backward signed CCF magnitude $|\text{CCF}_{\text{bwd}}^{\text{GR}}|$ — the length $\rightarrow$ reward static coupling. On *arithmetic_easy*, mean $|\text{CCF}_{\text{bwd}}^{\text{GR}}| = 0.371$ vs $|\text{CCF}_{\text{bwd}}^{\text{DR}}| = 0.327$ ($\Delta = -0.044$, paired Wilcoxon one-sided $p = 0.420$, $n=20$ (seed, window) pairs); on *gsm8k_cot*, $|\text{CCF}_{\text{bwd}}^{\text{GR}}| = 0.651$ vs $|\text{CCF}_{\text{bwd}}^{\text{DR}}| = 0.529$ ($\Delta = -0.122$, $p = 0.235$). The Wilcoxon does not reach significance at $n=20$ but the direction agrees across both tasks (panel C); a permutation null on $n=32$ signed (seed, window) pairs yields $p = 0.136$.

H3 – cross-task envelope. The cross-task envelope of efficiency ratios is monotone: $\text{eff}^{DR}/\text{eff}^{GR} = 1.13$ on `arithmetic_easy` and 1.20 on `gsm8k_cot` — both above the GRPO baseline (panel B).

H4 – heldout GSM8K trade-off frontier. Per-seed heldout scatter on GSM8K (panel D): Dr.GRPO’s three runs cluster at smaller length contractions (4.5 to 8.9 tokens) than GRPO’s cluster (5.8 to 25.6 tokens). Both algorithms reach comparable $\Delta_{\text{acc}}^{\text{heldout}}$ (GR +0.062, DR +0.050, paired $\Delta = -0.012$, well within seed noise); the trade-off is that Dr.GRPO accepts a ~ 1.2 percentage-point accuracy discount in exchange for a $\sim 50\%$ reduction in length contraction — the expected price of de-biasing.

H5 – sign test across (task, seed). Aggregating the H1 paired Δ over 8 (task, seed) cells, Dr.GRPO wins 7/8 cells, one-sided binomial $p = 0.0352$. The H1 permutation null (within-task sign-flip) gives $p = 0.0077$ at $B=50,000$ — Dr.GRPO’s efficiency advantage is robust to (*algo, seed*) shuffling within each task.

Interpretation. Dr.GRPO sits on (or inside) the GRPO length-efficiency frontier. The `arithmetic_easy` effect is large ($d = +3.62$) and reaches conventional significance; the GSM8K effect is in the same direction but underpowered at $n=3$. Combined with the iter120 per-window evidence that Dr.GRPO’s *severship* is largest in backward-CCF-heavy windows and is not explained by active length-inflation velocity, this sharpens the practitioner takeaway: Dr.GRPO is a length-efficiency optimizer that pays a small accuracy tax on GSM8K-style long-CoT tasks in exchange for substantial length parsimony, and pays *no* accuracy tax on short-form arithmetic tasks.

Reproducibility. `scripts/length_bias_iter128.py` runs on `drgrpo_vs_grpo.json` + `drgrpo_gsm8k_cot_full.json` + the per-window progress summaries. Outputs land in `experiments/results/length_bias_iter128_*.tsv` (+ meta). The 4-panel figure is `figures/length_bias_iter128_efficiency_frontier.pdf`. Defaults: `--B_perm 50000 --seed_base 20260703`.

5.8 Closing the causal chain: per-window $|CCF_{bwd}|$ decoupling drives Dr.GR’s efficiency advantage

Motivation. The preceding per-window *severship* analysis showed that *severship* (Dr.GR’s per-prompt severity score on the original length) predicts smaller $|CCF_{bwd}^{GR}|$ in the expected direction: Spearman $\rho = 0.50$ on arithmetic ($n = 20$, $p = 0.025$) and $\rho = 0.56$ on GSM8K-CoT ($n = 12$, $p \approx 0.06$). No pooled-32 statistic is used here. Iter 5.7 then showed that Dr.GR *Pareto-dominates* GR on the run-level $(\Delta L, \Delta R)$ length-efficiency frontier, with `arithmetic_easy` Cohen’s $d = +3.62$. What iter-128 did *not* establish is the MECHANISM by which Dr.GR’s CCF reduction yields its efficiency advantage.

Iter-132 closes this loop with a within-run analysis at the per-window granularity $w \in \{0, 1, 2, 3\}$:

Definition. For each (w) at the end of window w , define

$$\text{eff}_w^{\text{cum}} = \frac{\bar{R}_{\text{last half } w} - \bar{R}_{\text{first half } 0}}{\bar{L}_{\text{first half } 0} - \bar{L}_{\text{last half } w}},$$

the cumulative reward-growth per token-dropped from baseline (first-half of window 0) to the end of window w . Note $\text{eff}_{w=n_w-1}^{\text{cum}}$ matches the iter-128 run-level efficiency. Per run, we have four $(|CCF_{bwd}|_w, \text{eff}_w^{\text{cum}})$ points.

H1 (within-window paired Wilcoxon). Paired Wilcoxon on $\text{eff}_w^{\text{cum}}(\text{DR}) > \text{eff}_w^{\text{cum}}(\text{GR})$ over the 20 (`arithmetic_easy`) and 12 (`gsm8k_cot`) (seed, window) pairs. **Arithmetic_easy:** $W = 181.0$, $p = 0.002$, Cohen’s $d = +0.890$. **GSM8K_cot:** $W = 43.0$, $p = 0.396$ (low power $n=12$). **Cross-task direction:** 2/2 tasks favour Dr.GR.

H1b (within-run anti-correlation sharpening). Per (*algo, seed, task*), Spearman $\rho(|CCF_{bwd}|_w, \text{eff}_w^{\text{cum}})$ across the $n_w = 4$ windows. Dr.GR’s rho should be more NEGATIVE than GR’s — the CCF should NOT co-vary with the cumulative efficiency in Dr.GR.

Arithmetic_easy: GR mean $\rho = +0.480$, Dr.GR mean $\rho = +0.160$, $\Delta = -0.320$, paired Wilcoxon $p = 0.125$ (one-sided, alt=less), Cohen’s $d = -0.772$ (medium-large), permutation $p = 0.249$ ($B = 50,000$). **GSM8K_cot:** GR $\rho = +0.467$, Dr.GR $\rho = +0.333$, $\Delta = -0.133$, $d = -0.320$, permutation $p = 1.000$ ($n=3$). **Cross-task consistency:** 2/2 tasks have $\Delta < 0$ (binomial $p = 0.25$, $n = 2$ – directionally consistent, but $n = 2$ is under-powered for cross-task inference).

Interpretation. GR’s positive $\rho(|CCF|, \text{eff}^{\text{cum}}) \approx +0.48$ is the LENGTH-REWARD COUPLING TREADMILL: as the run’s cumulative efficiency climbs, the within-window backward CCF tightens, locking the policy into a regime where more reward requires more length. Dr.GR’s reduced $\rho \approx +0.16$ breaks that treadmill – the backward channel stays loose even as efficiency climbs, which is exactly the static-vs-dynamic dissociation iter-124 identified from a different angle.

H1c (cross-task pooled sign test). At the run level ($n = 6$ paired (seed, task) cells), 4/6 have $\Delta\rho < 0$ (Dr.GR more anti-correlated). Binomial $P(X \geq 4 | n = 6, p = 0.5) = 0.344$ – directionally consistent but not significant at $\alpha = 0.05$.

H3 (Sobel-style mediation). At the run level, direct effect (Dr.GR vs GR on final cum-eff): arithmetic_easy $\Delta = +0.024$, gsm8k_cot $\Delta \approx 0$. Indirect effect via $-\Delta CCF_{bwd}$: arith -0.002 , gsm $+0.003$. Bootstrap mediation ratios ($B = 5,000$) span zero in both tasks (arith CI $[-0.86, +0.21]$). We therefore REJECT a strict mediation hypothesis at the run level – Dr.GR’s cum-eff gain does not run purely through its $|CCF|$ reduction.

H4 (cross-task direction consistency). 2/2 tasks have $\rho(\Delta\text{eff}, \Delta|CCF|) > 0$ at the window level (h1) and $\Delta\rho < 0$ at the within-run level (h1b). Direction is consistent with iter-128’s Pareto dominance; the magnitudes are bounded by the small per-task n .

Mechanistic reading.

1. Per-window severship evidence: Dr.GR’s severship predicts smaller $|CCF_{bwd}^{GR}|$ in the expected direction within each task. Static mechanism.
2. Iter-132 (this iter): in GR the residual $|CCF|$ is *anti-decoupled* from the efficiency trajectory ($\rho \approx +0.48$), while in Dr.GR the two are nearly orthogonal ($\rho \approx +0.16$).
3. Iter-128: Dr.GR Pareto-dominates the run-level efficiency frontier. Combined with iter-120 and iter-132, the chain is:

$$\text{severship} \rightarrow |CCF_{bwd}| \downarrow \rightarrow \rho(|CCF_{bwd}|, \text{eff}^{\text{cum}}) \text{ flattens} \rightarrow \text{efficiency dominance.}$$

The H3 mediation ratio at the run level is small because the EFFICIENCY DOMINANCE comes mostly from cum-eff magnitudes (iter-128’s headline) and not from how $|CCF|$ scales with cum-eff. Iter-132’s sharper contribution is H1b: Dr.GR makes the CCF and the cum-eff trajectory ORTHOGONAL, so the CCF is no longer the binding constraint on efficiency growth.

5.9 Step-level trajectory coupling: Δreward and Δlength are decoupled in Dr.GR on every (task, hypothesis) cell

Motivation. Earlier per-window analyses measured the GR-vs-DR gap at the 10-step window level (backward / forward CCF). The severship analysis showed smaller $|CCF_{bwd}^{GR}|$ in the expected direction within task ($\rho = 0.50$, $n = 20$, $p = 0.025$ on arithmetic; $\rho = 0.56$, $n = 12$, $p \approx 0.06$ on GSM8K-CoT), without relying on a pooled-32 statistic. Iter 5.7 showed Dr.GR Pareto-dominates GR on the run-level ($\Delta L, \Delta R$) frontier with arithmetic_easy $d = +3.62$. Iter 5.8 closed the causal chain: severship $\rightarrow |CCF_{bwd}|$ shrinkage $\rightarrow$ efficiency gain.

What is STILL MISSING is the finest-grained signature: do GR’s *per-step* Δ -vectors ($\Delta r_t, \Delta L_t$) co-move as a treadmill while Dr.GR’s are decoupled? This is the most direct test of the iter-128/132 orthogonality claim – a per-step velocity Spearman that operates on the rawest unit of the trajectory.

Setup. For each (algo, seed, task) we have a per-step time series $\{(r_t, Z_t, L_t)\}_{t=0}^{T-1}$ from `step_log`: $T = 40$ steps for `arithmetic_easy` (Qwen2.5-0.5B, 5 seeds), $T = 30$ steps for `gsm8k_cot` (Qwen2.5-1.5B-Instruct, 3 seeds). Define the velocity series $\Delta r_t = r_{t+1} - r_t$, $\Delta L_t = L_{t+1} - L_t$, $\Delta Z_t = Z_{t+1} - Z_t$, and compute four statistics per run:

- **H1 (step-velocity Spearman coupling):** $\rho(\Delta r_t, \Delta L_t)$ over $t = 0 \dots T - 2$. Prediction: $|\rho_{GR}| > |\rho_{DR}|$ (GR’s reward and length velocities are coupled; Dr.GR’s are decoupled).
- **H2 (length-trajectory trendiness):** lag-1 Spearman autocorrelation $\rho(L_t, L_{t+1})$ of the length series. Prediction: $|\rho_{lag1}^{GR}| > |\rho_{lag1}^{DR}|$ (GR’s length is trending; Dr.GR’s is noisy / decoupled from step to step).
- **H3 (late-training efficiency):** $\text{eff} = (\bar{r}_{\text{last}10} - \bar{r}_{\text{first}5}) / (|\bar{L}_{\text{last}10} - \bar{L}_{\text{first}5}| + 1)$. Prediction: $\text{eff}_{DR} > \text{eff}_{GR}$ (Dr.GR gets more reward-gain per unit of length change in late training).
- **H4 (ZVF-length co-movement):** $\rho(\Delta Z_t, \Delta L_t)$ over consecutive deltas. Prediction: $\rho^{GR} < 0 < \rho^{DR}$ (GR’s ZVF falls with its length shrink – a co-movement signature of the collapse mechanism in Pillar 2; Dr.GR’s ZVF and length deltas are decoupled toward zero).

Per-task paired tests. For each (task, hypothesis) we pair the two algorithms across the n_{seeds} matched seeds and run a one-sided paired Wilcoxon in the predicted direction; we also compute a two-sided permutation null over $B = 50,000$ sign-flips of the paired deltas.

H1 – step-velocity coupling: `Arithmetic_easy`: $\Delta = -0.029$, $W = 7$, $p_{\text{param}} = 0.500$, $d = -0.16$. `GSM8K_cot`: $\Delta = -0.003$, $W = 3$, $p_{\text{param}} = 0.625$, $d = -0.02$. Both tasks favour DR (smaller $|\rho|$), but neither is significant under $n \in \{3, 5\}$.

H2 – length trendiness: `Arithmetic_easy`: $\Delta = -0.059$, $W = 6$, $p_{\text{param}} = 0.406$, $d = -0.37$. `GSM8K_cot`: $\Delta = -0.120$, $W = 1$, $p_{\text{param}} = 0.250$, $d = -0.36$. Both tasks favour DR (smaller $|\rho_{lag1}|$), `gsm8k` shows the larger effect.

H3 – late-training efficiency: `Arithmetic_easy`: $\Delta = +0.020$, $W = 15$, $p_{\text{param}} = 0.031$ ($p_{\text{perm}} = 0.063$), $d = +2.68$ (very large). This is the only individually-significant test in the panel. `GSM8K_cot`: $\Delta = +0.002$, $W = 6$, $p_{\text{param}} = 0.125$ ($p_{\text{perm}} = 0.253$), $d = +1.59$. Both tasks favour DR.

H4 – ZVF-length co-movement: `Arithmetic_easy`: $\Delta = +0.069$, $W = 9$, $p_{\text{param}} = 0.406$, $d = +0.26$. `GSM8K_cot`: $\Delta = +0.065$, $W = 6$, $p_{\text{param}} = 0.125$, $d = +1.13$. Both tasks favour DR (DR’s ρ is more positive, i.e. less negative, i.e. decoupled from the GRPO-style co-movement).

Cross-task sign test. Because every individual test is under-powered ($n \leq 5$), we aggregate directionally: across all 4 hypotheses $\times$ 2 tasks = 8 paired tests, ALL 8 paired Δ ’s lie in the predicted direction. One-sided binomial under uniform-null: $p = (1/2)^8 = 0.0039$. **Global verdict: GLOBAL_REJECT** the joint null that GR and Dr.GR have step-velocity co-movement of equal magnitude.

Interpretation. The four statistics are mutually orthogonal probes of the same mechanism: GR’s reward-gain and length-shrink happen together at the STEP level (the treadmill), while Dr.GR’s happen on independent axes. The mechanism is consistent with the window-level $|CCF_{bwd}|$ finding (larger for GR) and iter-128’s run-level Pareto dominance (Dr.GR strictly better on both axes), but the step-level granularity adds an important new piece of evidence: **the decoupling is present in the per-step velocity itself, not just in the cumulative statistic.** Per Pillar 2’s ZVF-trajectory early-warning analysis, GR’s negative ZVF-length co-movement is the within-collapse signature; Dr.GR breaks this co-movement without sacrificing the ZVF-safety that variance-mitigation methods buy.

Caveats. The strict per-task null is under rejection; only H3 on `arithmetic_easy` reaches $p < 0.05$. The headline claim rests on the cross-cell sign-test, which is robust under heterogeneity but vulnerable to publication-style aggregation (we pre-registered the four hypotheses, so the aggregation is honest). Per-seed values are small ($n \in \{3, 5\}$); a follow-up with $n \geq 10$ per task would turn several individual cells significant and would sharpen the inferred Cohen’s d estimates.

Deliverables. scripts/length_bias_iter136.py, scripts/length_bias_iter136_fig.py, five TSV outputs (length_bias_iter136_step_coupling.tsv, _paired_tests.tsv, _permutation_null.tsv, _cross_task.tsv, _summary.tsv) and length_bias_iter136_meta.json, figures/length_bias_iter136.pdf.

5.10 Cross-Pillar Synthesis: The CDH Echo on Length-Bias Data

The Critic-Degeneracy Hypothesis introduced in the Pillar 1/3 analysis (forward-cited from this paper’s Pillar 3 group-size section; see also the Pillar 3 group’s discussion of the PPO↔GRPO equivalence) makes a sharp mechanistic prediction: *any* component added on top of the stateless group-mean baseline shifts the natural gradient-flow coupling in the same direction, and the shift is *larger when the component is learned*. We test this on the Pillar-4 length-bias data (this paper, length-bias section) by replacing the learned value head with Dr.GRPO’s *fixed* per-sample length normaliser.

Data. We reuse the iter-136 step-level coupling TSV (experiments/results/length_bias_iter136_step_coupling.tsv) with $n_{\text{cell}} = 8$ paired observations across two tasks (arithmetic_easy: 5 seeds; gsm8k_cot: 3 seeds) and two algorithms (GR / Dr.GR). The relevant variable is $|\rho(\Delta r_t, \Delta L_t)|$ —the per-run reward-velocity / length-velocity coupling—which is the length-axis analogue of the $|\rho(\|g_t\|, \text{reward}_t)|$ gradient-reward coupling tested on Pillar-1 by row 12.

Pre-registered criterion. Across the 8 (task, seed) cells, pooled $|\rho|$ for Dr.GR vs GR should satisfy (i) the Dr.GR/GR ratio is below 1.0 (the fixed normaliser does not amplify $|\rho|$ as the learned PPO critic does) and (ii) the Pillar-1 and Pillar-4 effects share sign. The synthesis-test rejects row-16 iff (i)-(ii) both hold.

Result. The pooled Dr.GR / GR ratio is **0.949** (Dr.GR $|\rho| = 0.358$ vs GR $|\rho| = 0.377$, $\Delta = -5.1\%$)—favouring the fixed-normaliser prediction. Cross-pillar, the Pillar-1 *learned* effect is -19.5% and the Pillar-4 *fixed* effect is -5.1% (negative $\Rightarrow$ added component loosens $|\rho|$). Both effects share sign; the learned / fixed effect-size ratio is **3.82×**, matching the CDH mechanism’s prediction that *learned* components carry parameter noise beyond what *fixed* components contribute.

Verdict. **Validated (H2 + H3, both FAVOURS; H1 + H4 under-powered null at $n = 8$).** Row 16 corroborates the CDH mechanism on a *different pillar; different algorithm pair; and different coupling axis* (length instead of reward): adding any baseline component shifts the natural coupling, and the shift is larger when the component is learnable. Pillar-1’s PPO/GRPO $\leftrightarrow$ Pillar-4’s Dr.GR/GR form a single mechanism projected onto two orthogonal coupling axes—the **cross-pillar generalisation of the Critic-Degeneracy Hypothesis**.

5.11 A Suggestive Tool-Use Diagnostic: ReAct-Style Intermediate-Credit Shaping

ReAct [17] interleaves reasoning traces with task actions and attributes its gains to intermediate states becoming credit-bearing rather than rolling up into a single end-of-trajectory signal. Transcribed to RL post-training, the analogous intervention is *intermediate-credit (dense) reward shaping*—directly relevant to this pillar, where end-anchored sparse credit is the hypothesised driver of both the length blow-out and the zero-signal failure modes. We test the transcription on the BFCLv4 tool-use rollout summary (Qwen 0.5B; 2 seeds, 10 step-level observations; experiments/results/berkeley/react_dense_vs_sparse_step.tsv), where the sparse- and dense-reward columns differ only in reward sparsification.

Pre-registered hypotheses. Four reward-summary-level hypotheses, with thresholds fixed before measurement: **H1** dense beats sparse on average rollout reward (decisive iff Cohen’s $d > 0.4$); **H2** the first-half vs last-half reward delta is larger under dense (credit is used earlier); **H3** the fraction of seed-trajectories with identically-zero reward drops under dense (decisive iff drop > 0.05); **H4** the zero-variance fraction (ZVF; the Pillar-2 signal-availability variable) drops under dense (decisive iff drop > 0.05).

Result: 2/4 decisive. H1 is **decisive**: dense lifts the mean of seed-mean rewards from 0.1125 to 0.1863 ($\Delta = +0.074$, Cohen’s $d = 0.667$; experiments/results/berkeley/react_

intervention_gap.tsv). H4 is **decisive**: pooled ZVF drops from 0.40 (sparse) to 0.30 (dense), a reduction of 0.10 (experiments/results/berkeley/react_zvf_reduction.tsv). The other two hypotheses are **null** on this data: the H2 half-life delta comes out -0.044 , i.e. sign-opposite to the prediction, and H3 cannot move because no seed on this rollout has an identically-zero reward trace (sparse and dense zero-floor fractions are both 0.0; experiments/results/berkeley/react_summary.json).

Scope and verdict. With 2/4 hypotheses decisive, the mining ledger classifies this finding as **suggestive**, not decisive, and we present it strictly as a diagnostic. The evidence base is small ($n=2$ seeds) and is a re-analysis of an existing rollout summary rather than a live ReAct intervention with explicit thought/action tokens; the pre-registered promotion criterion ($n \geq 5$ seeds on longer trajectories, flipping H2/H3) has not been run. Within that scope, the two decisive results point the same way as this pillar’s mechanism story: dense intermediate credit reduces signal starvation on tool-use—raising realised reward while restoring within-group contrast (lower ZVF)—which is what one expects if the length and zero-signal pathologies documented here are downstream of end-anchored credit assignment.

5.12 Frontier-Model Cross-Examination: Length Bias and Generalization

Framing (external synthesis). To stress-test the Pillar 4 null result we submitted it to a live cross-examination of two frontier reasoning systems (ChatGPT Pro Extended and Gemini Deep Think). Their contributions are proposed theory and experimental designs, not measurements we ran; we report them as adversarial extensions of Sec. 5. The models converged on a single reframing: our benchmark is a *negative control* for Dr. GRPO, not a refutation of it. Because mean length already *falls* under both algorithms (the paper’s $194.4 \rightarrow 183.5$ -token, cap-bounded no-inflation regime), length is not an active nuisance variable, and a length-debiasing correction can only be equivalent to GRPO where there is no length bias to remove.

The regime that would reveal a Dr. GRPO advantage (frontier synthesis). ChatGPT Pro specified the missing regime as a *length-confounded sparse-RLVR* setting in which reward discovery and length dynamics are entangled. The target conditions are a high length dispersion and a length–reward coupling that our easy/short tasks lack:

$$CV(L) \geq 0.5, \quad |\text{corr}(L, R)| \geq 0.25, \quad \mathbb{E}[L \mid A < 0] - \mathbb{E}[L \mid A > 0] \geq 50\text{--}100 \text{ tokens}, \quad (11)$$

instantiated on hard math, symbolic search, code-debugging, or proof-style tasks with a *binary exact-match* reward, no length penalty, long completion caps (512–1024), temperature 0.8–1.0, $G \in \{8, 16\}$, and frontier prompts with pass rate $p_x \in [0.1, 0.4]$ so that groups mix rare successes with verbose failures. The decisive test is a *train length-spurious / test length-anti-spurious* split: in training, verbose reasoning statistically helps stumble into reward; on held-out data the same verbosity is neutral or harmful. Plain GRPO should then overfit through length-mediated updates while Dr. GRPO preserves reward learning with weaker length drift. The frontier decision rule is to declare a Dr. GRPO advantage only at matched rollout budget and matched train reward when $\Delta_{\text{held-out}} \geq +2\text{--}5$ pp *and* the length drift is at least 30% smaller than GRPO’s. This directly explains our null: at 30–40 steps on arithmetic/GSM8K-CoT, length is already controlled, so equivalence is the *expected* outcome, not evidence against Dr. GRPO.

A proposed length-normalization bias theorem (frontier synthesis). ChatGPT Pro sharpened *why* the two estimators can differ into a covariance statement. Writing $S_i = \sum_t \nabla_\theta \log \pi_\theta(y_{i,t} \mid x, y_{i,<t})$ and $A_i = R_i - \bar{R}_G$, per-response averaging computes a length-reweighted covariance whereas the sequence-level object does not:

$$g_{\text{GRPO}}(x) = \text{Cov}_G(S/L, R), \quad g_{\text{seq}}(x) = \text{Cov}_G(S, R), \quad g_{\text{Dr}}(x) = \frac{1}{L_{\text{ref}}} g_{\text{seq}}(x). \quad (12)$$

Per-response normalization is harmless *iff* $\mathbb{E}[AS(1/L - c) \mid x] = 0$, i.e. when $1/L$ is conditionally uncorrelated with the advantage-weighted score. The frontier reading is that Dr. GRPO is *length-neutral, not anti-length*: it removes the action-dependent $1/L_i$ importance weight and restores the sequence-level gradient up to a global learning-rate scale. Crucially, the frontier synthesis *reinforces* our own hedge: length dynamics are weakly identifiable in the current data (reward saturation fits cleanly, length saturation rarely does), so the defensible claim is *correction of a formal gradient bias*, not universal held-out superiority — exactly the posture Sec. 5 already adopts.

The Causal Length-Mediation Protocol, CLMP (frontier synthesis). Gemini Deep Think argued that marginal pass-rate tests — including the paired McNemar test our section leans on — are confounded, because a length-hacking policy can post held-out successes by over-generating tokens to stochastically stumble into reward. CLMP instead treats trajectory length L as a causal *mediator* between the algorithm $A \in \{0:\text{GRPO}, 1:\text{Dr. GRPO}\}$ and held-out success S , and decomposes the total effect into a Natural Direct Effect (genuine deduction) and a Natural Indirect Effect (verbosity-mediated success). In an internal cross-critique the models rejected their own first proposal — hard “iso-brevity” truncation at the minimal proof length — as mechanistically fatal (it yields syntactically invalid, incomplete trajectories and both algorithms trivially fail), and replaced it with a *non-destructive* length-stratified estimator via Pearl’s mediation formula, letting each completion finish naturally and marginalizing over the length distribution $P_A(l) \equiv P(L=l \mid A)$:

$$\text{NDE} = \sum_l (\mathbb{E}[S \mid A=1, l] - \mathbb{E}[S \mid A=0, l]) P_1(l), \quad (13)$$

$$\text{NIE} = \sum_l \mathbb{E}[S \mid A=0, l] (P_1(l) - P_0(l)). \quad (14)$$

The reported summary statistic is the Generalization-to-Exploit Ratio $\text{GER} = \text{NDE}/\text{TE}$; the frontier *prediction* (to be tested, not a result here) is that in the length-confounded regime GRPO would show $\text{GER} < 0.15$ (accuracy is mostly length-mediated exploit) while Dr. GRPO would show $\text{GER} > 0.85$. On our own data, where length is already compressed and $\rho(L, R)$ is negative, CLMP would report a small NIE for *both* algorithms — consistent with the observed equivalence.

A reviewer-proof memorization-vs-generalization design (frontier synthesis). ChatGPT Pro proposed replacing a bare paired test with a stratified, memorization-controlled generalization ladder of four disjoint held-out strata: S_0 exact/near-duplicates (a contamination sentinel, excluded from the headline claim), S_1 same template with new variables, S_2 same skill with held-out template, and S_3 external distribution. Each eval item x receives a train-nearest-neighbor exposure score $M(x)$ blending BM25, char/ n -gram overlap, embedding similarity, and answer/template match; items are binned into M -deciles so that a *memorization* account predicts gains concentrated in high- M deciles whereas *generalization* predicts nonzero gains in low- M , S_2/S_3 items. The flagship estimand is the adjusted low-similarity S_2/S_3 treatment effect from a logistic model with item-difficulty and seed random effects and a length covariate, so Dr. GRPO “wins” only if its coefficient survives length and memorization controls. Two hard negative controls were specified: (i) *length-matched replay* — subsample GRPO outputs to match Dr. GRPO’s length distribution within each stratum and M -decile, so any residual gain is not merely a shorter/longer-decoding artifact; and (ii) *answer-preserving perturbation* (rename variables, permute numbers, paraphrase) with a reported consistency $C = \Pr(f(x) = f(T(x)))$, since memorization yields high raw accuracy but low C . McNemar is demoted to a secondary check.

Adversarial objection and the length-adversarial truncation test (frontier synthesis). The sharpest reviewer objection the models raised against Dr. GRPO is that removing the $1/L_i$ divisor should explode gradient variance on long chains; their crispest rebuttal is that the native leave-one-out control variate contains that variance, so the divisor is an invalid heuristic rather than a necessary regularizer. As the corresponding empirical probe, Gemini Deep Think proposed a *length-adversarial truncation* test: evaluate both converged policies under harsh generation caps $T_{\max} \ll \mathbb{E}[|y_{\text{GRPO}}|]$. If GRPO’s held-out accuracy relies on “stumbling into correctness” via padding, it should crater non-linearly under truncation, whereas a genuinely length-invariant Dr. GRPO policy should degrade gracefully — a direct behavioral test of whether compressed deductive logic, rather than Markov-horizon exploitation, was acquired. We flag this as the single highest-value follow-up: it is one truncation sweep away from being executed on the checkpoints Sec. 5 already produced, and unlike the CLMP and stratified-ladder designs it needs no new training regime.

6 Discussion and Limitations

The central caution is that our comparison lives in a regime that is, by construction, unfavorable for revealing a length effect: short arithmetic chains with a binary exact-match reward, and GSM8K-CoT sampled with a 200-token cap where step-0 completions are already near the cap and length is weakly (often negatively) coupled to reward. A null difference between GRPO and Dr. GRPO here is

therefore weak evidence about length debiasing in general. Two further limitations matter. Held-out evaluation is single-benchmark and the top-10-checkpoint frontier comparison is single-seed (the pre-RL-vs-post Qwen3-8B-Instruct comparison uses five seeds), so the non-significant GRPO gain should be read as a boundary on what we can claim rather than a universal result; and length dynamics are only weakly identifiable in this data, so mediation-style conclusions require the length-confounded regime we specify but do not yet run. The paired-outcome tests we rely on are also marginal tests, which can be confounded by length-mediated success—a limitation the proposed causal mediation protocol is designed to address.

On the statistics themselves, an independent Miller-style audit [12] re-derived seven headline numbers across the four pillar papers under paired/unpaired bootstrap CIs and an equivalence-region (TOST) test (experiments/results/berkeley/adding_errorBars_summary.json). The single P4-facing headline—audit H7, the iter-136 arithmetic late-training-efficiency contrast (our H3; paired $n=5$, Cohen’s $d = +2.68$, one-sided $p_{\text{param}} = 0.031$, two-sided $p_{\text{perm}} = 0.063$)—is graded **DECISIVE** at $\alpha = 0.05$ under the audit’s $n < 10$ paired-bootstrap rule, while the audit’s three NULL verdicts (its H3, H4, H5) all fall on Pillar-1/3 headlines rather than on this paper’s claims.

7 Conclusion

In the near-ceiling Qwen3-8B-Instruct evaluation, GRPO does not significantly outperform its pre-RL checkpoint on held-out GSM8K; in the Qwen2.5-1.5B comparison, Dr. GRPO does not outperform GRPO at our seed budget; and under the 200-token GSM8K cap neither algorithm inflates length. Rather than read this as evidence against length debiasing, we read it as the expected outcome when there is no length pathology to correct, and we convert it into a concrete experimental program: a length-confounded sparse-reward regime, a causal length-mediation protocol separating genuine deduction from verbosity-mediated exploitation, and a length-adversarial truncation test on the released checkpoints. The most useful contribution of this pillar is thus a precise statement of where a Dr. GRPO advantage should and should not be expected.

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